

Potential-Function Proofs for Methods of the Gradient Descent Framework

Lola (Leona) Rapoport

March 2023

*School of Computer Science and Engineering
The Hebrew University of Jerusalem*

Abstract

This note discusses simple proofs of convergence for a variety of gradient methods (i.e., *first-order methods*) using potential functions. We begin by describing known results of basic gradient descent and projected gradient descent (for both smooth and non-smooth settings). We consider for non-smooth settings both the online case and the offline case and mention that there is a better regret (and convergence) for strongly convex functions. We then describe the Frank-Wolfe method (also called *conditional gradient descent*) for smooth functions and prove the error is $O(1/T)$, where T is the number of iterations. We also discuss gradient descent for well-conditioned functions. Next, we cover the results of basic mirror descent and show proofs for gradient descent for smooth functions and well-conditioned functions with respect to general norms instead of the Euclidean norm. Finally, we describe Nesterov's accelerated gradient descent (both for the smooth case and for the well-conditioned case) and provide proofs for both cases. We also show for the smooth case proof for arbitrary norms.

Contents

1	Introduction	3
1.1	Basic Definitions and Properties	3
1.2	The Problems and Assumptions	4
1.3	Results and Further Work	5
2	Online Analyses	7
2.1	Basic Gradient Descent	7
2.1.1	Projected Gradient Descent	8
2.2	Strong Convexity Analysis	9

3	Bounds for Smooth Functions	11
3.1	Smooth Gradient Descent	11
3.1.1	Yet Another Bound	12
3.1.2	Projected Smooth Gradient Descent	12
3.1.3	The Frank-Wolfe Method	12
3.2	Well-Conditioned Functions	15
4	The Mirror-Descent Framework	19
4.1	Basic Mirror Descent	19
4.1.1	The Update Rules	20
4.1.2	Bregman Divergence	21
4.1.3	The Analysis	23
4.2	A Parenthesis: Smooth Functions and General Norms	25
4.3	A Parenthesis: Well-Conditioned Functions and General Norms	27
5	Nesterov Acceleration: A Potential Function Proof	30
5.1	Accelerated Gradient Descent	30
5.1.1	A Parenthesis: Optimizing Parameters and Making Connections	32
5.2	The Constrained Case With Acceleration	34
5.3	The Extension to Arbitrary Norms	37
5.4	Acceleration for Strongly Convex Functions	41
A	Appendix	48
A.1	A Failed Attempt to Prove Theorem 3.12	48

1 Introduction

The gradient descent framework consists of iterative methods that are commonly used for solving convex minimization problems. The idea of moving in the direction opposite to the gradient naturally arises, since the gradient indicates the direction of steepest increase in function value. Over the years, different variants have developed from this general idea which is central to convex optimization. In recent years, since the approach of using discrete optimization methods has become popular, the above idea has also become central to designing algorithms in general. In this note, we cover convergence arguments of Bansal and Gupta [1] for various commonly used versions of gradient methods (also known as *first-order methods*) using simple *potential-function* proofs. We also present proofs for convergence arguments that were not fully proven in [1] and show proof for well-conditioned functions with respect to arbitrary norms instead of the Euclidean norm.

1.1 Basic Definitions and Properties

Fact 1.1. $\forall a, b \in \mathbb{R}^n$ the following identity holds:

$$\|a + b\|^2 = \|a\|^2 + 2\langle a, b \rangle + \|b\|^2,$$

where the norm $\|\cdot\|$ is the Euclidean norm.

Fact 1.2. $\forall a, b, c \in \mathbb{R}^n$ the following identity holds:

$$\|a + b + c\|^2 = \|a\|^2 + \|b\|^2 + \|c\|^2 + 2\langle a, b \rangle + 2\langle b, c \rangle + 2\langle a, c \rangle,$$

where the norm $\|\cdot\|$ is the Euclidean norm.

Definition 1.3 (Convex Set). A set $K \subseteq \mathbb{R}^n$ is called *convex* if for all $x, y \in K$,

$$\lambda x + (1 - \lambda)y \in K, \tag{1.1}$$

for all values of $\lambda \in [0, 1]$. Geometrically, this means that for any two points in K , the line connecting them is contained in K .

Definition 1.4 (Convex Function). A function $f : K \rightarrow \mathbb{R}$ defined on a convex set K is called *convex* if for all $x, y \in K$, any of the following holds:

1. (Zeroth order) $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $\lambda \in [0, 1]$.
2. (First order) If f is differentiable, then

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle. \tag{1.2}$$

3. (Second order) If f is twice-differentiable, then its Hessian matrix $\nabla^2 f(x)$ is positive-semidefinite at every point $x \in K$. In other words, all eigenvalues of $\nabla^2 f(x)$ are non-negative at every point $x \in K$.

Definition 1.5 (Projection). Given a convex set $K \subseteq \mathbb{R}^n$, the *projection* of point $\tilde{b} \in \mathbb{R}^n$ onto K is $\Pi_K(\tilde{b}) := \arg \min_{b \in K} \|b - \tilde{b}\|$.

Proposition 1.6 (Pythagorean Property). *Given a convex set $K \subseteq \mathbb{R}^n$, let $a \in K$ and $\tilde{b} \in \mathbb{R}^n$. Let $b = \Pi_K(\tilde{b})$. Then $\langle a - b, \tilde{b} - b \rangle \leq 0$ and thus $\|a - b\|^2 \leq \|a - \tilde{b}\|^2$.*

Proof. For the first part, the separating hyperplane at b has all of K (and hence a) on one side and $\tilde{b}$ on the other side. Therefore, the angle between $\tilde{b} - b$ and $a - b$ must be obtuse and as a result $\langle a - b, \tilde{b} - b \rangle \leq 0$. Hence, $\underbrace{\langle a - b, \tilde{b} - b \rangle}_{=\langle a - b, \tilde{b} - b \rangle} \geq$

0. For the second part, according to Fact 1.1, we have $\|a - \tilde{b}\|^2 = \|a - b\|^2 + \|b - \tilde{b}\|^2 + 2\langle a - b, b - \tilde{b} \rangle$. Since the latter two terms are non-negative, the inequality $\|a - b\|^2 \leq \|a - \tilde{b}\|^2$ holds. ■

1.2 The Problems and Assumptions

As mentioned above, gradient methods are well-known methods in the context of convex optimization and we consider the following defined types of problems:

Definition 1.7 (Unconstrained Convex Minimization Problem). Given a convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and an error parameter ε , the *unconstrained convex minimization* problem is to find a point $\hat{x} \in \mathbb{R}^n$ such that $f(\hat{x}) - \min_{x \in \mathbb{R}^n} f(x) \leq \varepsilon$.

Definition 1.8 (Constrained Convex Minimization Problem). Given a convex function $f : K \rightarrow \mathbb{R}$ defined on a convex domain $K \subseteq \mathbb{R}^n$ and an error parameter ε , the *constrained convex minimization* problem is to find a point $\hat{x} \in K$ such that $f(\hat{x}) - \min_{x \in K} f(x) \leq \varepsilon$.

In both types of problems, x^* denotes the minimizer of f . Our goal will be to bound the number of gradient queries required to converge to the approximate minimizer $\hat{x}$, as a function of the distance between x_0 and x^* and some parameters of the function f , determined by the variant of gradient descent. We also discuss the following general type of problem:

Definition 1.9 (Online Convex Minimization Problem). Given a convex set $K \subseteq \mathbb{R}^n$, at each timestep $t = 1, 2, \dots$, the algorithm proposes a point $x_t \in K$ and an adversary gives a convex function $f_t : K \rightarrow \mathbb{R}$. The algorithm's *loss* at timestep t is defined to be $f_t(x_t)$, so that in total the *regret* of the algorithm is $\sum_{t=1}^T f_t(x_t) - \min_{x \in K} \sum_{t=1}^T f_t(x)$. The objective is to minimize the regret by determining the points x_t online, i.e., without knowing the current or future functions $\{f_s\}_{s \geq t}$.

Note that online convex optimization generalizes the two types of problems mentioned above, which corresponds to setting $f_t = f$ at each timestep t , and that any algorithm with sublinear regret $o(T)$ can be used to find an approximate minimizer $\hat{x}$ up to any desired accuracy ε .

For this note, we assume our convex sets K are closed with non-empty interior and that our convex functions are closed, convex and differentiable. However, it is possible to extend the proofs to non-differentiable functions easily using subgradients. In detail, all we need is the following:

Definition 1.10 (Subgradient). A vector z_x is called a *subgradient* at point x if

$$f(y) \geq f(x) + \langle z_x, y - x \rangle$$

for all $y \in \mathbb{R}^n$.

Now we can use subgradients at point x wherever we used $\nabla f(x)$, and the entire proof goes through. In fact, in some cases an approximate subgradient may be sufficient. We also assume access to a gradient oracle for each of the functions f_t we consider, that is, given any point x , it is possible to get the gradient $\nabla f_t(x)$ of function f_t at point x . We only consider the Euclidean norm $\|\cdot\|_2$ for the first few sections. General norms are discussed in the last two sections.

1.3 Results and Further Work

We use the same pattern for the potential in all proofs in the offline minimization setting: for some fixed point x^* (which we can consider as the optimal or reference point) we have

$$\boxed{\phi_t = a_t \cdot (f(x_t) - f(x^*)) + b_t \cdot (\text{distance from } x_t \text{ to } x^*)}, \quad (1.3)$$

where a_t, b_t are non-negative multipliers. Note that the distance functions may vary. The approach here is to show that $\phi_{t+1} - \phi_t \leq B_t$, where in many cases B_t is zero for each timestep t . Since the potential and distance terms are non-negative, the telescoping sum $\sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} B_t$ gives

$$\begin{aligned} \phi_T &\leq \phi_0 + \sum_{t=0}^{T-1} B_t \\ \Rightarrow f(x_T) - f(x^*) &\leq \frac{\phi_0 + \sum_{t=0}^{T-1} B_t}{a_T}. \end{aligned}$$

Section 2 contains the results of basic gradient descent and projected gradient descent (for both smooth and non-smooth settings). These results can be extended to the online regret-minimization setting, where the function may change at each timestep. Here, the analysis is more similar to *amortized analysis*: the

approach is to show that the amortized cost, which is the cost of the algorithm plus the increase in potential is at most the optimal cost (plus some constant B). Namely, $f_t(x_t) + (\phi_{t+1} - \phi_t) \leq f_t(x^*) + B$. Since $\phi_{T+1} \geq 0$, the telescoping sum

$$\begin{aligned} \sum_{t=1}^T (f_t(x_t) + (\phi_{t+1} - \phi_t)) &\leq \sum_{t=1}^T (f_t(x^*) + B) \\ \Rightarrow \sum_{t=1}^T (f_t(x_t) - f_t(x^*)) &\leq \sum_{t=1}^T (\phi_t - \phi_{t+1}) + \underbrace{\sum_{t=1}^T B}_{=T \cdot B} \end{aligned}$$

gives us that the average regret is:

$$\frac{1}{T} \left(\sum_{t=1}^T (f_t(x_t) - f_t(x^*)) \right) \leq B + \frac{\phi_1}{T}.$$

An improvement of both the regret bound and the convergence bound for strongly convex functions is discussed as well.

Next, in Section 3, we mention several convergence arguments of both gradient descent and projected gradient descent for *smooth* convex functions. In addition, we introduce the Frank-Wolfe method and the error is $O(1/T)$, where T is the number of iterations. We also discuss gradient descent for well-conditioned functions.

We then present *mirror descent*, which is a generalization of gradient descent to arbitrary norms. The potential used in the proof of the regret bound consists of Bregman divergence, which is a natural generalization of the squared Euclidean norm. Moreover, we show proofs for gradient descent for smooth functions and well-conditioned functions with respect to general norms using Fenchel duality. Section 4 contains these proofs.

Finally, in Section 5, we give potential-function proofs of Nesterov's accelerated gradient descent method both for the smooth case and for the well-conditioned case and show for the smooth case a full proof for general norms.

Two directions for further work can be considered. Namely, one can try to give a potential-function proof for the well-conditioned case, in the constrained case. Another essential direction is to show whether it is possible to extend the proof of Nesterov's accelerated gradient descent given for the well-conditioned case to arbitrary norms.

2 Online Analyses

2.1 Basic Gradient Descent

In [1, 2] the results of basic gradient descent and projected gradient descent (for both smooth and non-smooth settings) are presented. The analyses of both methods work even for the online convex optimization setting: at each timestep t we receive a function f_t , we select x_t and the objective is to minimize the regret. In this case the update rule is:

$$\boxed{x_{t+1} \leftarrow x_t - \eta_t \nabla f_t(x_t)}, \quad (2.1)$$

where x_0 is the starting point and η_t is the step size at time t . By taking derivatives with respect to x , we can obtain an equivalent form for this update rule:

$$\boxed{x_{t+1} \leftarrow \arg \min_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - x_t\|^2 + \eta_t \langle x, \nabla f_t(x_t) \rangle \right\}}. \quad (2.2)$$

The main idea is simple: since the gradient indicates the direction of steepest increase, we would like to move opposite to the direction of the gradient to decrease the fastest.

In basic gradient descent, the step size is often fixed for all times t (i.e., $\eta_t = \eta$ for all times t), so the algorithm performs the update rule (2.1) for some number of steps T , and then returns the average of all the points seen during the process. Also, in this note, in some methods (namely gradient descent, projected gradient descent and mirror descent), the point x_1 has the value of the starting point x_0 .

Theorem 2.1 (Basic Gradient Descent). *Let $f_1, \dots, f_T : \mathbb{R}^n \rightarrow \mathbb{R}$ be G -Lipschitz functions, i.e. $\|f_t(x)\| \leq G$ for all x, t . Let x_0 be the starting point. Then using the update rule (2.1) with step size*

$$\eta_t = \eta = \frac{D}{G\sqrt{T}}$$

for T steps guarantees an average regret of

$$\frac{1}{T} \sum_{t=1}^T (f_t(x_t) - f_t(x^*)) \leq \eta \frac{G^2}{2} + \frac{D^2}{2\eta T} \leq \frac{DG}{\sqrt{T}},$$

for all x^ with $\|x_0 - x^*\| \leq D$.*

This regret bound can be shown using the following potential function

$$\boxed{\phi_t = \frac{1}{2\eta} \|x_t - x^*\|^2}. \quad (2.3)$$

A complete proof is given in [1]. Note that we can obtain from this regret bound a convergence result for the offline case, i.e., for the case where $f_t = f$ for all t . Below is the algorithm of basic gradient descent for the offline case:

Algorithm 1 Basic Gradient Descent for the Offline Case

```

1:  $x_1 \leftarrow$  starting point
2: for  $t \leftarrow 1$  to  $T$  do
3:    $x_{t+1} \leftarrow x_t - \eta_t \cdot \nabla f(x_t)$ 
4: end for
5: return  $\hat{x} := \frac{1}{T} \sum_{t=1}^T x_t$ .
```

Theorem 2.2 (Basic Gradient Descent: Offline Optimization). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a G -Lipschitz function, i.e. $\|f(x)\| \leq G$ for all x . Let x_0 be the starting point. If we define $T \geq \left(\frac{DG}{\varepsilon}\right)^2$ and $\eta_t = \frac{\varepsilon}{G^2}$ for all steps, then the solution $\hat{x}$ returned by basic gradient descent satisfies*

$$f(\hat{x}) \leq f(x^*) + \varepsilon. \quad (2.4)$$

Proof. By definition of $\hat{x}$, the convexity of f and Theorem 2.1,

$$f(\hat{x}) - f(x^*) = f\left(\frac{1}{T} \sum_{t=1}^T x_t\right) - f(x^*) \leq \frac{1}{T} \sum_{t=1}^T (f(x_t) - f(x^*)) \leq \frac{DG}{\sqrt{T}} \leq \varepsilon,$$

where the last inequality holds since $T \geq \left(\frac{DG}{\varepsilon}\right)^2$. ■

It is important to note that we can obtain a convergence result even if the time horizon T is unknown, by setting a time-dependent step size of $\eta_t = \frac{D}{G\sqrt{t}}$ and using an identical proof. It is also well-known that the convergence bound above is the best possible in general modulo constant factors (see, for example, [4] or [5]).

2.1.1 Projected Gradient Descent

If we want to solve the constrained minimization problem for a convex set $K \subseteq \mathbb{R}^n$, we update as follows:

$$\boxed{\begin{aligned} \tilde{x}_{t+1} &\leftarrow x_t - \eta_t \nabla f_t(x_t), \\ x_{t+1} &\leftarrow \Pi_K(\tilde{x}_{t+1}). \end{aligned}} \quad (2.5)$$

Basically, to ensure that the new point x_{t+1} lies within K , we simply project the new iterate back onto K . This is illustrated in Figure 1.

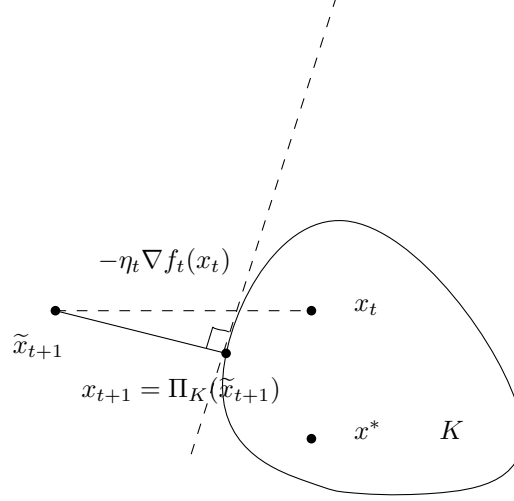


Figure 1: Projected Gradient Descent.

The modified algorithm for the offline case is given below in Algorithm 2.

Algorithm 2 Projected Gradient Descent for the Offline Case

- 1: $x_1 \leftarrow$ starting point
 - 2: **for** $t \leftarrow 1$ to T **do**
 - 3: $\tilde{x}_{t+1} \leftarrow x_t - \eta_t \cdot \nabla f(x_t)$
 - 4: $x_{t+1} \leftarrow \Pi_K(\tilde{x}_{t+1})$
 - 5: **end for**
 - 6: **return** $\hat{x} := \frac{1}{T} \sum_{t=1}^T x_t$.
-

Using 1.6, we get that for any point $x^* \in K$,

$$\|x_{t+1} - x^*\|^2 \leq \|\tilde{x}_{t+1} - x^*\|.$$

Using this inequality and following the analysis in the proof of Theorem 2.1 with the same potential function (2.3), we can observe that the projection only helps and we receive an identical regret bound (and thus we can obtain the same convergence result as in Theorem 2.2).

2.2 Strong Convexity Analysis

A better regret (and convergence) bound can be obtained when the functions are “not too flat”, i.e. when the function are strongly convex.

Definition 2.3 (Strong Convexity). A function $f : K \rightarrow \mathbb{R}$ over a convex set $K \subseteq \mathbb{R}^n$ is α -strongly convex if for all $x, y \in K$, any of the following holds:

1. (Zeroth order) $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) - \frac{\alpha}{2}\lambda(1-\lambda)\|x-y\|^2$ for all $\lambda \in [0, 1]$.
2. (First order) If f is differentiable, then
$$f(y) \geq f(x) + \langle \nabla f(x), y-x \rangle + \frac{\alpha}{2}\|x-y\|^2. \quad (2.6)$$
3. (Second order) If f is twice-differentiable, then all eigenvalues of $\nabla^2 f(x)$ are at least α at every point $x \in K$.

For α -strongly convex functions f_t , we use the same update rule but vary the step size η_t . In detail,

$$x_{t+1} \leftarrow \Pi_K(x_t - \eta_t \nabla f_t(x_t)), \quad (2.7)$$

where $\eta_t = \frac{1}{\alpha(t+1)}$. In this setting, the total regret only increases logarithmically as $\log T$ with the time, whereas it increases as $\sqrt{T}$ in Theorem 2.1.

Theorem 2.4 (Strongly Convex Functions [1]). *If the functions f_t are α -strongly convex and G is an upper bound on $\|f_t(x)\|$ for all $x \in K, t \in \{1, \dots, T\}$ the update rule (2.7) with $\eta_t = \frac{1}{\alpha(t+1)}$ guarantees an average regret of*

$$\frac{1}{T} \sum_{t=1}^T (f_t(x_t) - f_t(x^*)) \leq \frac{G^2 \log T}{2T\alpha}.$$

This regret bound can be shown using the following potential function

$$\phi_t = \frac{1}{2\eta_{t-1}} \|x_t - x^*\|^2 = \frac{t\alpha}{2} \|x_t - x^*\|^2.$$

It turns out that this bound of $O(\log T)$ on the average regret is tight, as Takimoto and Warmuth [3] showed a matching lower bound. However, in the *offline* optimization setting where there is a fixed function $f_t = f$, we can achieve a convergence rate of $O(\frac{1}{T})$ with respect to a convex combination of the points x_t .

Theorem 2.5 (Strongly Convex Functions: Offline Optimization [1]). *Let f be α -strongly convex and G is the upper bound on $\|f(x)\|$ for all x , and let x_t be the iterates produced by the update rule (2.1) with $\eta_t = \frac{1}{\alpha t}$. For any $T \geq 1$, let $\bar{x}_T := \sum_{t=1}^T \lambda_t x_t$ denote the convex combination of x_t with*

$$\lambda_t = \frac{2t}{T(T+1)}.$$

Then,

$$f(\bar{x}_T) - f(x^*) \leq \frac{G^2}{\alpha(T+1)}.$$

Note that we can extend Theorem 2.5 to the constrained case using the same analysis as in the proof with the same potential function, exactly for the same reason as in Section 2.1.1.

3 Bounds for Smooth Functions

Assuming that the functions are “not too curved”, i.e. they are Lipschitz smooth, the average regret cannot be improved, as the average regret of $O(1/\sqrt{T})$ is tight even for linear functions. However, a better convergence rate can be obtained for the *offline* setting, where the function $f_t = f$ for all time steps.

Definition 3.1 (Lipschitz Smoothness). A function $f : K \rightarrow \mathbb{R}$ over a convex set $K \subseteq \mathbb{R}^n$ is β -Lipschitz smooth if for all $x, y \in K$, any of the following holds:

1. (Zeroth order) $f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y) - \frac{\beta}{2}\lambda(1-\lambda)\|x-y\|^2$ for all $\lambda \in [0, 1]$.
2. (First order) If f is differentiable, then

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|x - y\|^2. \quad (3.1)$$

3. (Second order) If f is twice-differentiable, then all eigenvalues of $\nabla^2 f(x)$ are at most β at every point $x \in K$.

The smoothness condition is equivalent to requiring that the gradients are Lipschitz conditions, that is, $\|\nabla f(x) - \nabla f(y)\| \leq \beta \|x - y\|$ for all x, y .

3.1 Smooth Gradient Descent

In this case, the update rule has a time-invariant multiplier ($\eta_t := \frac{1}{\beta}$):

$$\boxed{x_{t+1} \leftarrow x_t - \frac{1}{\beta} \nabla f(x_t)}. \quad (3.2)$$

We can get a sub-optimal convergence bound with an additional $\log T$ factor using the following very natural potential function

$$\boxed{\phi_t = t \cdot (f(x_t) - f(x^*))}.$$

Theorem 3.2 (Smooth Functions [1]). *If f is β -smooth and $D := \max_x \{\|x - x^*\|_2 \mid f(x) \leq f(x_0)\}$, the update rule (3.2) guarantees*

$$f(x_T) - f(x^*) \leq \beta \frac{D^2(1 + \ln T)}{2T}.$$

This bound can be improved by modifying slightly the potential. Namely, the logarithmic dependence in Theorem 3.2 can be removed by multiplying the potential used in the proof by a linear term in t . So using instead the potential function

$$\boxed{\phi_t = t(t+1) \cdot (f(x_t) - f(x^*))}$$

gives the next better convergence bound.

Theorem 3.3 (Smooth Functions: Take II [1]). *If f is β -smooth, and $D := \max_x \{\|x - x^*\|_2 \mid f(x) \leq f(x_0)\}$, the update rule (3.2) guarantees*

$$f(x_T) - f(x^*) \leq \beta \frac{2D^2}{T+1}.$$

3.1.1 Yet Another Bound

Another bound without the logarithmic term can be achieved:

Theorem 3.4 (Smooth Functions: Take III [1]). *If f is β -smooth, the update rule (3.2) guarantees*

$$f(x_T) - f(x^*) \leq \beta \frac{\|x_0 - x^*\|^2}{2T}.$$

This bound is almost the same as in Theorem 3.3, with a slightly better definition of the distance term (D^2 against $\|x_0 - x^*\|^2$). This convergence bound can be achieved using the following potential function

$$\boxed{\phi_t = t \cdot (f(x_t) - f(x^*)) + \frac{a}{2} \|x_t - x^*\|^2},$$

where a is determined to be β , according to the analysis in the proof of Theorem 3.4.

3.1.2 Projected Smooth Gradient Descent

Let's consider the constrained minimization problem for a given convex set $K \subseteq \mathbb{R}^n$. As before, the update rule includes a step taken and a projection back onto K :

$$\boxed{\begin{aligned} \tilde{x}_{t+1} &\leftarrow x_t - \frac{1}{\beta} \nabla f(x_t), \\ x_{t+1} &\leftarrow \Pi_K(\tilde{x}_{t+1}). \end{aligned}} \quad (3.3)$$

Using the same potential function as in Theorem 3.4 gives the following convergence bound for the constrained case.

Theorem 3.5 (Constrained Smooth Optimization). *If f is β -smooth, the update rule (3.3) guarantees*

$$f(x_T) - f(x^*) \leq \frac{\beta}{2} \frac{\|x_0 - x^*\|^2}{T}.$$

3.1.3 The Frank-Wolfe Method

One disadvantage of projected gradient descent is the projection step: given a point $\tilde{x}$ and a convex set $K \subseteq \mathbb{R}^n$, finding the closest point $\Pi_K(\tilde{x})$ might be computationally expensive. Alternatively, we can use a different update rule, the *Frank-Wolfe* method (also known as *conditional gradient descent*),

that implements each gradient step using linear optimization over the set K . Basically, at each timestep we find the point in K that is furthest from the current point in the direction opposite to the gradient, and move a small distance towards it. This is illustrated in Figure 2.

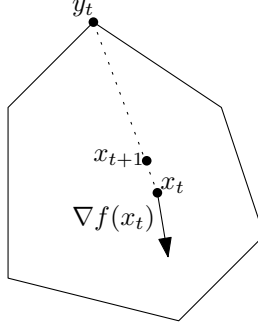


Figure 2: The Frank-Wolfe Update.

So, the update rule for the Frank-Wolfe method is simply as follows:

$$\begin{aligned} y_t &\leftarrow \arg \min_{y \in K} \langle \nabla f(x_t), y \rangle, \\ x_{t+1} &\leftarrow (1 - \eta_t)x_t + \eta_t y_t. \end{aligned} \quad (3.4)$$

Setting $\eta_t = \frac{2}{t+2}$ in advance will give the following result:

Theorem 3.6 (Smooth-Functions: Frank-Wolfe). *If f is β -smooth, K is a convex set with diameter $D := \max_{x,y \in K} \|x - y\|$, then the update rule (3.4) with $\eta_t = \frac{2}{t+2}$ ¹ guarantees*

$$f(x_T) - f(x^*) \leq 2\beta \frac{D^2}{T+1}.$$

Proof. Consider the following potential

$$\phi_t = t(t+1) \cdot (f(x_t) - f(x^*)).$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (t+1)(t+2) \cdot (f(x_{t+1}) - f(x^*)) - t(t+1) \cdot (f(x_t) - f(x^*)) \\ &= (t+1)(t+2) \cdot (f(x_{t+1}) - f(x_t)) + 2(t+1) \cdot (f(x_t) - f(x^*)). \end{aligned} \quad (3.5)$$

¹Originally in [1] it was stated that setting $\eta_t = \frac{2}{t+1}$ with the same potential function gives the result. However, I believe this is an error, since in my proof in which the same potential function is used I only achieved the result by setting $\eta_t = \frac{2}{t+2}$.

To bound the change in potential (3.5), note that by the update rule (3.4) the equality $x_{t+1} - x_t = \eta_t(y_t - x_t)$ holds. Thus,

$$\begin{aligned}
f(x_{t+1}) - f(x_t) &\stackrel{\text{by smoothness}}{\leq} \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 \\
&= \eta_t \langle \nabla f(x_t), y_t - x_t \rangle + \frac{\beta \eta_t^2}{2} \|y_t - x_t\|^2 \\
&\stackrel{\text{by optimality of } y_t}{\leq} \eta_t \langle \nabla f(x_t), x^* - x_t \rangle + \frac{\beta \eta_t^2}{2} \|y_t - x_t\|^2
\end{aligned}$$

Also, by convexity of f , we have $f(x_t) - f(x^*) \leq \langle \nabla f(x_t), x_t - x^* \rangle$. Putting it all together, we have

$$\begin{aligned}
\phi_{t+1} - \phi_t &\leq (t+1)(t+2) \cdot \left(\eta_t \langle \nabla f(x_t), x^* - x_t \rangle + \frac{\beta \eta_t^2}{2} \|y_t - x_t\|^2 \right) \\
&\quad + 2(t+1) \cdot \langle \nabla f(x_t), x_t - x^* \rangle \\
&= (t+1)(\eta_t(t+2) - 2) \langle \nabla f(x_t), x^* - x_t \rangle + (t+1)(t+2) \cdot \frac{\beta \eta_t^2}{2} \|y_t - x_t\|^2
\end{aligned}$$

Setting $\eta_t := \frac{2}{t+2}$ cancels the linear terms and hence

$$\phi_{t+1} - \phi_t \leq (t+1)(t+2) \frac{\beta}{2} \cdot \frac{4}{(t+2)^2} D^2 = 2\beta D^2 \frac{t+1}{t+2} \leq 2\beta D^2,$$

where the last inequality holds since $\frac{t+1}{t+2} \leq 1$ for all t . By summing over t and using $\phi_0 = 0$, we have

$$\phi_T = \phi_T - \phi_0 = \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 2\beta D^2 = 2\beta D^2 \cdot T.$$

Hence, by definition of the potential:

$$f(x_T) - f(x^*) = \frac{\phi_T}{T(T+1)} \leq \frac{2\beta D^2 \cdot T}{T(T+1)} = 2\beta \frac{D^2}{T+1}.$$

Hence the proof. ■

We can obtain a convergence bound with logarithmic dependence using the same potential as in Theorem 3.2, which is obtained from this potential by omitting the second linear term (i.e., $\phi_t = t \cdot (f(x_t) - f(x^*))$), but with $\eta_t = \frac{1}{t+1}$. Apart from these, the proof remains the same.

Theorem 3.7 (Smooth-Functions: Frank-Wolfe, Take II). *If f is β -smooth, K is a convex set with diameter $D := \max_{x,y \in K} \|x - y\|$, then the update rule (3.4) with $\eta_t = \frac{1}{t+1}$ guarantees*

$$f(x_T) - f(x^*) \leq \beta \frac{D^2(1 + \ln T)}{2T}.$$

3.2 Well-Conditioned Functions

If a function f is both α -strongly convex and β -smooth, it must be that $\alpha \leq \beta$. Thus, the eigenvalues of the Hessian $\nabla^2 f(x)$ must lie in the interval $[a, b]$ at all points $x \in K$. The ratio $\kappa := \frac{\beta}{\alpha}$ is called the *condition number*. Functions with small κ values are called *well-conditioned functions*. For such functions, we can achieve ε -closeness in time $T = \Theta(\log \frac{1}{\varepsilon})$, where the constant depends on the condition number $\kappa = \frac{\beta}{\alpha}$. This convergence rate is called *linear convergence* in the numeric analysis literature. The update rule is the same as for smooth functions:

$$x_{t+1} \leftarrow x_t - \frac{1}{\beta} \nabla f(x_t). \quad (3.6)$$

We first consider the unconstrained case and show this stronger convergence guarantee using a potential function, but also simply using the definitions of smoothness and strong convexity. We start by showing the convergence guarantee without using a potential function.

Theorem 3.8 (Well-Conditioned Optimization: Take I). *Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that is both α -strongly convex and β -smooth, define $\kappa := \beta/\alpha$. Let x^* be the solution to the unconstrained problem $\arg \min_{x \in \mathbb{R}^n} f(x)$. The update rule (3.6) ensures*

$$f(x_T) - f(x^*) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*)).$$

Proof. We assume $\kappa > 1$, since $\kappa = 1 \Leftrightarrow f(x) = ax^T x + b^T x + c$ for suitable scalars a, c and $b \in \mathbb{R}^n$. In this case, it is easily checked that the optimum solution x^* is achieved in a single step. Since f is a function that is β -smooth, we can use its smoothness with $x = x_T$ and $y = x_{T+1} = x_T - \frac{1}{\beta} \nabla f(x_T)$ to get:

$$\begin{aligned} f(x_{T+1}) - f(x_T) &\leq \langle \nabla f(x_T), x_{T+1} - x_T \rangle + \frac{\beta}{2} \|x_{T+1} - x_T\|^2 = \left\langle \nabla f(x_T), -\frac{1}{\beta} \nabla f(x_T) \right\rangle \\ &\quad + \frac{\beta}{2} \left\| -\frac{1}{\beta} \nabla f(x_T) \right\|^2 = -\frac{1}{\beta} \overbrace{\langle \nabla f(x_T), \nabla f(x_T) \rangle}^{=\|\nabla f(x_T)\|^2} + \frac{\beta}{2} \cdot \frac{1}{\beta^2} \|\nabla f(x_T)\|^2 \\ &= \left(-\frac{1}{\beta} + \frac{1}{2\beta} \right) \cdot \|\nabla f(x_T)\|^2 = -\frac{1}{2\beta} \|\nabla f(x_T)\|^2. \end{aligned} \quad (3.7)$$

Since f is a function that is α -strongly convex, we can use its strong convexity with $x = x_T$ and $y = x^*$ to get:

$$f(x_T) - f(x^*) \leq \langle \nabla f(x_T), x_T - x^* \rangle - \frac{\alpha}{2} \|x_T - x^*\|^2 \leq \frac{1}{2\alpha} \|\nabla f(x_T)\|^2, \quad (3.8)$$

where the second inequality uses $\langle a, b \rangle - \frac{\|b\|^2}{2} \leq \frac{\|a\|^2}{2}$ with $a = \frac{1}{\sqrt{\alpha}} \nabla f(x_T)$ and $b = \sqrt{\alpha}(x_T - x^*)$. Therefore, $-2\alpha \cdot (f(x_T) - f(x^*)) \geq -\|\nabla f(x_T)\|^2$ and combining this with 3.7 gives

$$f(x_{T+1}) - f(x_T) \leq -\frac{\alpha}{\beta} \cdot (f(x_T) - f(x^*)) = -\frac{1}{\kappa} \cdot (f(x_T) - f(x^*)). \quad (3.9)$$

Since x^* is the optimal solution to the unconstrained problem, we have $f(v) \geq f(x^*) = \min_{x \in \mathbb{R}^n} f(x)$ for all $v \in \mathbb{R}^n$. So in particular, we have $f(x_T) \geq f(x^*)$ and hence $f(x_T) - f(x^*) \geq 0$. Adding this to (3.9) and rearranging, we get

$$\begin{aligned} f(x_{T+1}) - f(x^*) &\leq \left(1 - \frac{1}{\kappa}\right) \cdot (f(x_T) - f(x^*)) \\ &\leq \left(1 - \frac{1}{\kappa}\right)^T \cdot (f(x_1) - f(x^*)) \leq \left(1 - \frac{1}{\kappa}\right)^{T+1} \cdot (f(x_0) - f(x^*)). \end{aligned}$$

Therefore, dividing by $(1 - \frac{1}{\kappa})$ (which $\neq 0$ by our assumption about κ) gives

$$f(x_T) - f(x^*) \leq \left(1 - \frac{1}{\kappa}\right)^T \cdot (f(x_0) - f(x^*)) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*)).$$

Hence the proof. $\blacksquare$

Theorem 3.9 (Well-Conditioned Optimization: Take II). *Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that is both α -strongly convex and β -smooth, define $\kappa := \beta/\alpha$. The update rule (3.6) ensures*

$$f(x_T) - f(x^*) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*)) \quad (3.10)$$

for all x^* .

Proof. Again, we assume $\kappa > 1$ for simplicity for the same reason mentioned above. We define $\gamma = \frac{1}{\kappa-1}$ and use the potential

$$\boxed{\phi_t = (1 + \gamma)^t \cdot (f(x_t) - f(x^*))}. \quad (3.11)$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (1 + \gamma)^{t+1} \cdot (f(x_{t+1}) - f(x^*)) - (1 + \gamma)^t \cdot (f(x_t) - f(x^*)) \\ &= (1 + \gamma)^t \cdot ((1 + \gamma)(f(x_{t+1}) - f(x_t)) + \gamma(f(x_t) - f(x^*))). \end{aligned} \quad (3.12)$$

To bound the first term, we use the smoothness of f with $x = x_t$ and $y = x_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$:

$$\begin{aligned} f(x_{t+1}) - f(x_t) &\leq \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 = \left\langle \nabla f(x_t), -\frac{1}{\beta} \nabla f(x_t) \right\rangle \\ &\quad + \frac{\beta}{2} \left\| -\frac{1}{\beta} \nabla f(x_t) \right\|^2 = -\frac{1}{\beta} \overbrace{\langle \nabla f(x_t), \nabla f(x_t) \rangle}^{= \|\nabla f(x_t)\|^2} + \frac{\beta}{2} \cdot \frac{1}{\beta^2} \|\nabla f(x_t)\|^2 \\ &= \left(-\frac{1}{\beta} + \frac{1}{2\beta} \right) \cdot \|\nabla f(x_t)\|^2 = -\frac{1}{2\beta} \|\nabla f(x_t)\|^2. \end{aligned}$$

To bound the second term, we use the strong convexity of f with $x = x_t$ and $y = x^*$:

$$f(x_t) - f(x^*) \leq \langle \nabla f(x_t), x_t - x^* \rangle - \frac{\alpha}{2} \|x_t - x^*\|^2 \leq \frac{1}{2\alpha} \|\nabla f(x_t)\|^2,$$

where the second inequality uses $\langle a, b \rangle - \frac{\|b\|^2}{2} \leq \frac{\|a\|^2}{2}$ with $a = \frac{1}{\sqrt{\alpha}} \nabla f(x_t)$ and $b = \sqrt{\alpha}(x_t - x^*)$. Plugging those bounds back into (3.12) gives:

$$\phi_{t+1} - \phi_t \leq (1 + \gamma)^t \cdot \left(-\frac{1 + \gamma}{2\beta} + \frac{\gamma}{2\alpha} \right) \|\nabla f(x_t)\|^2.$$

Since $\gamma = \frac{1}{\kappa - 1} = \frac{\frac{\beta}{\alpha}}{\frac{\beta}{\alpha} - 1}$, we have

$$\begin{aligned} -\frac{1 + \gamma}{2\beta} + \frac{\gamma}{2\alpha} &= \frac{-\alpha(1 + \gamma) + \beta\gamma}{2\beta\alpha} = \frac{-\alpha \left(1 + \frac{1}{\beta/\alpha - 1} \right) + \frac{\beta}{\beta/\alpha - 1}}{2\beta\alpha} \\ &= \frac{-\alpha + \frac{\beta - \alpha}{1/\alpha(\beta - \alpha)}}{2\beta\alpha} = \frac{-\alpha + \frac{1}{1/\alpha}}{2\beta\alpha} = \frac{\overbrace{-\alpha + \alpha}^{=0}}{2\beta\alpha} = 0 \end{aligned}$$

Thus, $\phi_{t+1} - \phi_t \leq 0$ for all t and after T steps

$$\begin{aligned} \phi_T - \phi_0 &= \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 0 = 0 \\ &\Rightarrow \phi_T \leq \phi_0 \end{aligned}$$

So by definition of ϕ_t , we have

$$\begin{aligned} (1 + \gamma)^T (f(x_T) - f(x^*)) &\leq (1 + \gamma)^0 (f(x_0) - f(x^*)) = f(x_0) - f(x^*) \\ &\Rightarrow f(x_T) - f(x^*) \leq (1 + \gamma)^{-T} (f(x_0) - f(x^*)) \\ &= \left(1 + \frac{1}{\kappa - 1} \right)^{-T} (f(x_0) - f(x^*)) = \left(\frac{\kappa - 1}{\kappa - 1} + \frac{1}{\kappa - 1} \right)^{-T} (f(x_0) - f(x^*)) \\ &= \left(\frac{\kappa}{\kappa - 1} \right)^{-T} (f(x_0) - f(x^*)) = \left(\frac{\kappa - 1}{\kappa} \right)^T (f(x_0) - f(x^*)) \\ &= \left(1 - \frac{1}{\kappa} \right)^T (f(x_0) - f(x^*)) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*)). \end{aligned}$$

Hence the proof. ■

Note 3.10. The use of the potential function gave us a more general convergence rate, which is not exclusive to $x^* = \arg \min_{x \in \mathbb{R}^n} f(x)$, but valid for all $x^* \in \mathbb{R}^n$.

Now we can show that the algorithm's point x_T also gets quickly closer to x^* . If x^* is the minimizer of f , we know $\nabla f(x^*) = \vec{0}$. Hence, smoothness gives us $f(x_0) - f(x^*) \leq \frac{\beta}{2} \|x_0 - x^*\|^2$, and strong convexity gives $f(x_T) - f(x^*) \geq \frac{\alpha}{2} \|x_T - x^*\|^2$. Plugging those inequities into (3.10) gives us

$$\begin{aligned} \frac{\alpha}{2} \|x_T - x^*\|^2 &\leq \exp(-T/\kappa) \cdot \frac{\beta}{2} \|x_0 - x^*\|^2 \\ &\Rightarrow \|x_T - x^*\|^2 \leq \frac{\beta}{\alpha} \exp(-T/\kappa) \cdot \|x_0 - x^*\|^2 \\ &\Rightarrow \|x_T - x^*\|^2 \leq \kappa \exp(-T/\kappa) \cdot \|x_0 - x^*\|^2. \end{aligned}$$

As for the constrained case, the update rule involves taking a step and then projecting back onto K :

$$\boxed{\begin{aligned}\tilde{x}_{t+1} &\leftarrow x_t - \frac{1}{\beta} \nabla f(x_t), \\ x_{t+1} &\leftarrow \Pi_K(\tilde{x}_{t+1}).\end{aligned}} \quad (3.13)$$

According to [1], in order to show this stronger convergence guarantee, the following claim should be used:

Claim 3.11. If f is both an α -strongly convex and a β -smooth function over a convex set $K \subseteq \mathbb{R}^n$, then for any $y \in K$,

$$f(x_{t+1}) - f(y) \leq \beta \langle x_t - x_{t+1}, x_t - y \rangle - \frac{\beta}{2} \|x_t - x_{t+1}\|^2 - \frac{\alpha}{2} \|x_t - y\|^2.$$

Proof. For any $y \in K$, we have $f(x_{t+1}) - f(y) = (f(x_{t+1}) - f(x_t)) + (f(x_t) - f(y))$. Now using smoothness and strong convexity for the first and second terms respectively, we have

$$\begin{aligned}f(x_{t+1}) - f(y) &\leq \left(\langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 \right) \\ &\quad + \left(\langle \nabla f(x_t), x_t - y \rangle - \frac{\alpha}{2} \|x_t - y\|^2 \right) \\ &= \langle \nabla f(x_t), x_{t+1} - y \rangle + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 - \frac{\alpha}{2} \|x_t - y\|^2. \quad (3.14)\end{aligned}$$

Since $\beta \langle x_{t+1} - \tilde{x}_{t+1}, x_{t+1} - y \rangle \leq 0$ according to Proposition 1.6, we have $\beta \langle \tilde{x}_{t+1} - x_{t+1}, x_{t+1} - y \rangle \geq 0$ and thus

$$\begin{aligned}\langle \nabla f(x_t), x_{t+1} - y \rangle &\stackrel{(3.13)}{=} \langle \beta(x_t - \tilde{x}_{t+1}), x_{t+1} - y \rangle \leq \beta \langle x_t - \tilde{x}_{t+1}, x_{t+1} - y \rangle \\ &\quad + \beta \langle \tilde{x}_{t+1} - x_{t+1}, x_{t+1} - y \rangle = \beta \langle x_t - x_{t+1}, x_{t+1} - y \rangle \\ &= \beta \langle x_t - x_{t+1}, x_{t+1} - y - x_t + x_t \rangle = \beta \langle x_t - x_{t+1}, x_t - y \rangle - \beta \underbrace{\|x_{t+1} - x_t\|^2}_{= \langle x_t - x_{t+1}, x_t - x_{t+1} \rangle}\end{aligned}$$

By substituting into (3.14) we get

$$\begin{aligned}f(x_{t+1}) - f(y) &\leq \beta \langle x_t - x_{t+1}, x_t - y \rangle - \beta \|x_{t+1} - x_t\|^2 + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 \\ &\quad - \frac{\alpha}{2} \|x_t - y\|^2 = \beta \langle x_t - x_{t+1}, x_t - y \rangle - \frac{\beta}{2} \|x_{t+1} - x_t\|^2 - \frac{\alpha}{2} \|x_t - y\|^2 \\ &= \beta \langle x_t - x_{t+1}, x_t - y \rangle - \frac{\beta}{2} \|x_t - x_{t+1}\|^2 - \frac{\alpha}{2} \|x_t - y\|^2\end{aligned}$$

■

Theorem 3.12 (GD: Constrained Well-Conditioned Optimization). *Given a function $f : K \rightarrow \mathbb{R}$ over a convex set $K \subseteq \mathbb{R}^n$ that is both α -strongly convex and β -smooth, define $\kappa := \beta/\alpha$. The update rule (3.13) ensures*

$$f(x_T) - f(x^*) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*))$$

for all x^* .

Unfortunately, I have not yet been able to prove the theorem using Claim 3.11 and the same potential function used in the proof of Theorem 3.9, as mentioned in [1]. In Appendix A.1, a failed attempt to prove the theorem is described. However, there are proofs without the use of a potential function for the constrained case in which a stronger convergence rate is achieved (see, for example, [4, Thm 3.10] or [7, Thm 2.6]).

One can ask whether the convergence for smooth and for well-conditioned functions is optimal as a function of T . The answer is no, and there is a known result of Nesterov which gives faster (and optimal) convergence rates. We show this result (which can be proved using a potential-function based proof) in Section 5.

4 The Mirror-Descent Framework

4.1 Basic Mirror Descent

The main concept of mirror descent is to define an injective mapping between E to E^* , which is called the *mirror map*. Given a point x_t , we first map it to E^* , make the gradient step there, and then use the inverse map back to receive the point x_{t+1} . Let's start with some basic definitions and notation. Let E be some vector space with an inner product $\langle \cdot, \cdot \rangle$, and define a norm $\|\cdot\|$ on E . To measure distances in E^* , the *dual norm* is used which is defined as follows:

Definition 4.1 (Dual Norm). Let $\|\cdot\|$ be a norm. The dual norm of $\|\cdot\|$ is a function $\|\cdot\|_*$ defined as

$$\|y\|_* := \sup\{\langle x, y \rangle : \|x\| \leq 1\}.$$

- The dual norm of the ℓ_2 -norm is again the ℓ_2 -norm. In other words, the Euclidean norm is self-dual.
- The dual norm for the ℓ_p -norm is the ℓ_q -norm, where $\frac{1}{p} + \frac{1}{q} = 1$.

Corollary 4.2 (Cauchy-Schwartz for General Norms). *For $x, y \in \mathbb{R}^n$, we have $\langle x, y \rangle \leq \|x\| \|y\|_*$.*

Proof. Assume $\|x\| \neq 0$, otherwise both sides are 0 and the inequality holds. Since $\|x/\|x\|\| = 1$ and by definition of the dual norm, we have $\langle x/\|x\|, y \rangle \leq \|y\|_*$. Multiplying both sides of this inequality by $\|x\|$ gives the required inequality. ■

To define a mirror map, we first fix a norm $\|\cdot\|$, and then choose a differentiable convex function $h : \mathbb{R}^n \rightarrow \mathbb{R}$ that is α -strongly convex with respect to this norm. Such a function must satisfy

$$h(y) \geq h(x) + \langle \nabla h(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$$

and it defines a map from E to E^* via its gradient (the map $x \mapsto \nabla h(x)$). The strong convexity of h ensures the map is injective (i.e., $\nabla h(x) = \nabla h(y)$ implies $x = y$). In addition, this map is also surjective, so for any point $\theta \in E^*$ there is an inverse $x \in E$ such that $\nabla h(x) = \theta$. This inverse map is actually given by the gradient of the Fenchel dual for h , that is, $\nabla h^*(\theta) = x \Leftrightarrow \nabla h(x) = \theta$ (it suffices to just construe $\nabla h^*(\theta)$ as $\nabla h^{-1}(\theta)$).

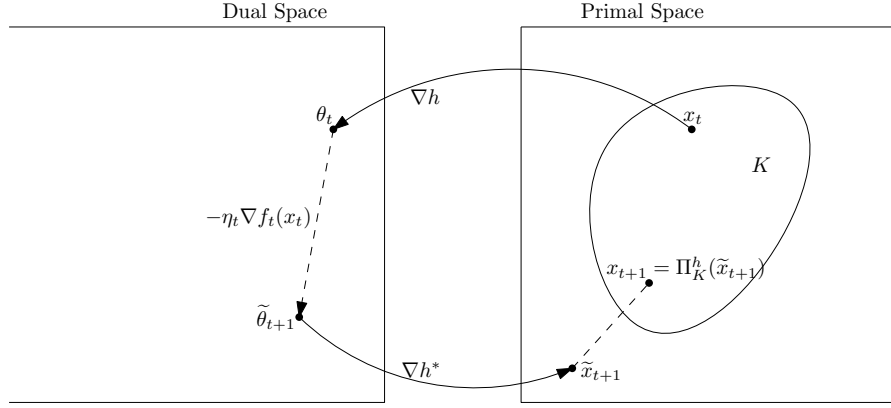


Figure 3: Mirror Descent.

4.1.1 The Update Rules

Here, we also keep track of the image point $\theta_t = \nabla h(x_t)$, since the map $x \mapsto \nabla h(x)$ is a bijective map from the primal space E to the dual space E^* . Naturally the update rules are given by

$$\begin{aligned} \tilde{\theta}_{t+1} &\leftarrow \theta_t - \eta_t \nabla f_t(x_t), \\ \tilde{x}_{t+1} &\leftarrow \nabla h^*(\tilde{\theta}_{t+1}), \\ x_{t+1} &\leftarrow \arg \min_{x \in K} D_h(x \| \tilde{x}_{t+1}). \end{aligned} \tag{4.1}$$

Note that we do not want to refer only to the Euclidean norm but to any arbitrary norm. Thus, instead of using the Euclidean norm as a distance for the projection, we use the *Bregman divergence* $D_h(y \| x)$, which we define later.

Alternatively, the mirror descent update can be presented as:

$$x_{t+1} \leftarrow \arg \min_{x \in K} \{ \langle \eta_t \nabla f_t(x_t), x - x_t \rangle + D_h(x \| x_t) \}. \tag{4.2}$$

This is a generalization of (2.2). This can be easily shown for the unconstrained case by taking derivatives. For the constrained case, KKT conditions are used.

4.1.2 Bregman Divergence

Given a strictly convex function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, we can define a distance based on how the function differs from its linear approximation (the convexity of h ensures that this quantity is non-negative):

Definition 4.3 (Bregman Divergence). The *Bregman divergence* from x to y with respect to function h is

$$D_h(y||x) := h(y) - h(x) - \langle \nabla h(x), y - x \rangle.$$

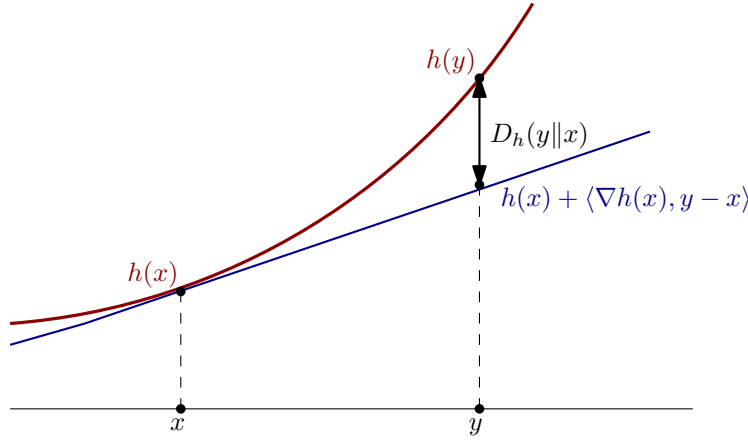


Figure 4: $D_h(y||x)$ for Function $h : \mathbb{R} \rightarrow \mathbb{R}$.

Figure 4 illustrates this definition geometrically for a function of one variable $h : \mathbb{R} \rightarrow \mathbb{R}$. Note that if h is α -strongly convex with respect to the norm $\|\cdot\|$, then $D_h(y||x) \geq \frac{\alpha}{2} \|y - x\|^2$. Also, $D_h(y||x)$ is strictly convex in its first argument, but not necessarily in the second argument. The gradient of the divergence with respect to the first argument is $\nabla_y(D_h(y||x)) = \nabla h(y) - \nabla h(x)$. Here are a few examples:

- The function $h(x) = \frac{1}{2} \|x\|_2^2$ from $\mathbb{R}^n$ to $\mathbb{R}$ is 1-strongly convex with respect to the Euclidean norm (and hence strictly convex), and the associated Bregman divergence is $D_h(y||x) = \frac{1}{2} \|y - x\|_2^2$, half the squared Euclidean distance.
- For the negative entropy function $h(x) = \sum_{i=1}^n x_i \ln x_i$ defined on the probability simplex $\Delta_n := \{x \in [0, 1]^n \mid \sum_{i=1}^n x_i = 1\}$. For $x, y \in \Delta_n$, the associated Bregman distance $D_h(y||x)$ is $\sum_{i=1}^n y_i \ln(\frac{y_i}{x_i})$, the relative entropy or *Kullback-Leibler (KL) divergence* from x to y .

Definition 4.4 (Bregman Projection). Given a convex set $K \subseteq \mathbb{R}^n$ and a strictly convex function h , the *Bregman projection* of a point $\tilde{x}$ on K is

$$\Pi_K^h(\tilde{x}) = \arg \min_{x \in K} D_h(x \| \tilde{x}).$$

Note that if $\tilde{x} \in K$, then $\Pi_K^h(\tilde{x}) = \tilde{x}$ since $D_h(\tilde{x} \| \tilde{x}) = 0$. Also, a very vital property of the Bregman projections is that they satisfy a “Pythagorean inequality” with respect to the divergence (which is analogous to Proposition 1.6).

Proposition 4.5 (Generalized Pythagorean Property). *Given a convex set $K \subseteq \mathbb{R}^n$, let $a \in K$ and $\tilde{b} \in \mathbb{R}^n$. Let $b = \Pi_K^h(\tilde{b})$. Then*

$$\langle \nabla h(\tilde{b}) - \nabla h(b), a - b \rangle \leq 0.$$

In particular,

$$D_h(a \| \tilde{b}) \geq D_h(a \| b) + D_h(b \| \tilde{b}), \quad (4.3)$$

and hence $D_h(a \| b) \leq D_h(a \| \tilde{b})$.

Proof. For the first part, recall that for any convex function g and a convex set $K \subseteq \mathbb{R}^n$, if $x^* = \arg \min_{x \in K} g(x)$ is the minimizer of g in K , then $\langle \nabla g(x^*), y - x^* \rangle \geq 0$ for all $y \in K$. Consider $g(x) = D_h(x \| \tilde{b})$ and note that $g(x)$ is convex with $\nabla g(x) = \nabla h(x) - \nabla h(\tilde{b})$ and that $x^* = b$ is the minimizer. Thus, $\langle \nabla h(b) - \nabla h(\tilde{b}), a - b \rangle \geq 0$ for all $a \in K$, which means $\langle \nabla h(\tilde{b}) - \nabla h(b), a - b \rangle \leq 0$ for all $a \in K$. For the second part, we use the last inequality and expand the terms:

$$\begin{aligned} \langle \nabla h(\tilde{b}) - \nabla h(b), a - b \rangle &\leq 0 \Rightarrow -\langle \nabla h(\tilde{b}), a - b + \tilde{b} - \tilde{b} \rangle \geq -\langle \nabla h(b), a - b \rangle \\ &\Rightarrow -\langle \nabla h(\tilde{b}), a - \tilde{b} \rangle \geq -\langle \nabla h(b), a - b \rangle - \langle \nabla h(\tilde{b}), b - \tilde{b} \rangle \\ &\Rightarrow \underbrace{h(a) - h(\tilde{b}) - \langle \nabla h(\tilde{b}), a - \tilde{b} \rangle}_{=D_h(a \| \tilde{b})} \geq \underbrace{h(a) - h(b) - \langle \nabla h(b), a - b \rangle}_{=D_h(a \| b)} \\ &\quad + \underbrace{h(b) - h(\tilde{b}) - \langle \nabla h(\tilde{b}), b - \tilde{b} \rangle}_{=D_h(b \| \tilde{b})} \\ &\Rightarrow D_h(a \| \tilde{b}) \geq D_h(a \| b) + D_h(b \| \tilde{b}). \end{aligned}$$

Now, we use that the divergences are non-negative, so in particular $D_h(b \| \tilde{b}) \geq 0$ and hence

$$D_h(a \| \tilde{b}) \geq D_h(a \| b) + D_h(b \| \tilde{b}) \geq D_h(a \| b) + 0 = D_h(a \| b).$$

■

4.1.3 The Analysis

We consider the more general *online* optimization setting, and in particular the mirror descent algorithm described in Algorithm 3.

Algorithm 3 Mirror Descent for the Online Case

- 1: **Require:** $f_1, \dots, f_T$ convex functions defined on a convex set $K \subseteq \mathbb{R}^n$, a norm $\|\cdot\|$ and a distance-generating function $h : \mathbb{R}^n \leftarrow \mathbb{R}$, which gives the mirror map $\nabla h : \mathbb{R}^n \leftarrow \mathbb{R}^n$.
 - 2: $x_1 \leftarrow$ starting point
 - 3: **for** $t \leftarrow 1$ to T **do**
 - 4: $\theta_t \leftarrow \nabla h(x_t)$ ▷ mapping to the dual space
 - 5: $\tilde{\theta}_{t+1} \leftarrow \theta_t - \eta_t \cdot \nabla f_t(x_t)$ ▷ taking a gradient step in the dual space
 - 6: $\tilde{x}_{t+1} \leftarrow \nabla h^*(\tilde{\theta}_{t+1})$ ▷ mapping $\tilde{\theta}_{t+1}$ back to the primal space
 - 7: $x_{t+1} \leftarrow \arg \min_{x \in K} D_h(x \| \tilde{x}_{t+1})$ ▷ projecting $\tilde{x}_{t+1}$ back into K
 - 8: **end for**
-

We prove the following regret bound for this algorithm:

Theorem 4.6 (Mirror Descent Regret Bound). *Let $K \subseteq \mathbb{R}^n$ be a convex set, $f_1, \dots, f_T$ be convex functions defined on K , $\|\cdot\|$ be a norm, and h be an α_h -strongly convex function with respect to $\|\cdot\|$. The mirror descent algorithm starting at x_0 and taking constant step size $\eta_t = \eta$ in every iteration, produces $x_1, \dots, x_T$ such that*

$$\sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x^*) \leq \frac{D_h(x^* \| x_0)}{\eta} + \frac{\eta \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2}{2\alpha_h},$$

for all $x^* \in K$.

Proof. Consider the following potential

$$\boxed{\phi_t = \frac{D_h(x^* \| x_t)}{\eta}}. \tag{4.4}$$

Note that choosing $h(x) = \frac{1}{2} \|x\|_2^2$ gives us the potential function for the Euclidean norm (2.3). The potential change is

$$\phi_{t+1} - \phi_t = \frac{D_h(x^* \| x_{t+1})}{\eta} - \frac{D_h(x^* \| x_t)}{\eta} = \frac{1}{\eta} (D_h(x^* \| x_{t+1}) - D_h(x^* \| x_t))$$

and we have

$$\begin{aligned}
& D_h(x^* \| x_{t+1}) - D_h(x^* \| x_t) \\
& \quad \underbrace{\text{by generalized}}_{\text{Pythagorean Property}} \leq D_h(x^* \| \tilde{x}_{t+1}) - D_h(x^* \| x_t) \\
& = h(x^*) - h(\tilde{x}_{t+1}) - \underbrace{\left\langle \nabla h(\tilde{x}_{t+1}), x^* - \tilde{x}_{t+1} \right\rangle}_{=\tilde{\theta}_{t+1}} - h(x^*) + h(x_t) \\
& \quad + \underbrace{\left\langle \nabla h(x_t), x^* - x_t \right\rangle}_{=\theta_t} - \langle \nabla h(\tilde{x}_{t+1}), x_t - x^* \rangle + \underbrace{\langle \nabla h(\tilde{x}_{t+1}), x_t - x^* \rangle}_{=-\langle \tilde{\theta}_{t+1} x^* - x_t \rangle} \\
& = h(x_t) - h(\tilde{x}_{t+1}) - \underbrace{\left\langle \underbrace{\tilde{\theta}_{t+1}}_{=\theta_t - \eta_t \nabla f_t(x_t)}, x_t - \tilde{x}_{t+1} \right\rangle}_{=\theta_t - \eta_t \nabla f_t(x_t)} - \underbrace{\left\langle \underbrace{\tilde{\theta}_{t+1} - \theta_t}_{=-\eta_t \nabla f_t(x_t)}, x^* - x_t \right\rangle}_{=-\eta_t \nabla f_t(x_t)} \\
& \quad \underbrace{h(x_t) - h(\tilde{x}_{t+1}) - \langle \theta_t, x_t - \tilde{x}_{t+1} \rangle}_{\leq (-\alpha_h/2) \|x_t - \tilde{x}_{t+1}\|^2 \text{ by strong convexity of } h} + \langle \eta_t \nabla f_t(x_t), x_t - \tilde{x}_{t+1} \rangle + \langle \eta_t \nabla f_t(x_t), x^* - x_t \rangle \\
& \leq -\frac{\alpha_h}{2} \|x_t - \tilde{x}_{t+1}\|^2 + \eta_t \langle \nabla f_t(x_t), x_t - \tilde{x}_{t+1} \rangle + \eta_t \langle \nabla f_t(x_t), x^* - x_t \rangle \\
& = -\frac{\alpha_h}{2} \|x_t - \tilde{x}_{t+1}\|^2 + \eta_t \langle x_t - \tilde{x}_{t+1}, \nabla f_t(x_t) \rangle + \eta_t \langle \nabla f_t(x_t), x^* - x_t \rangle \\
& \leq \frac{\eta_t^2}{2\alpha_h} \|\nabla f_t(x_t)\|_*^2 + \eta_t \langle \nabla f_t(x_t), x^* - x_t \rangle. \quad (4.5)
\end{aligned}$$

The last inequality uses generalized Cauchy-Schwartz to get $\langle a, b \rangle \leq \|a\| \cdot \|b\|_* \leq \|a\|^2/2 + \|b\|^2/2$ with $a = \sqrt{\alpha_h}(x_t - \tilde{x}_{t+1})$ and $b = \frac{\nabla f_t(x_t)}{\sqrt{\alpha_h}}$. Recall that we set $\eta_t = \eta$ for all steps. Hence, dividing (4.5) and substituting result in

$$\phi_{t+1} - \phi_t \leq \frac{\eta}{2\alpha_h} \|\nabla f_t(x_t)\|_*^2 + \langle \nabla f_t(x_t), x^* - x_t \rangle$$

and thus

$$\begin{aligned}
f_t(x_t) - f_t(x^*) + (\phi_{t+1} - \phi_t) & \leq \underbrace{f_t(x_t) - f_t(x^*) + \langle \nabla f_t(x_t), x^* - x_t \rangle}_{\leq 0 \text{ by convexity of } f_t} \\
& \quad + \frac{\eta}{2\alpha_h} \|\nabla f_t(x_t)\|_*^2 \leq 0 + \frac{\eta}{2\alpha_h} \|\nabla f_t(x_t)\|_*^2 = \frac{\eta}{2\alpha_h} \|\nabla f_t(x_t)\|_*^2.
\end{aligned}$$

Summing over all steps to get the total regret,

$$\begin{aligned}
\sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x^*) &\leq \sum_{t=1}^T (\phi_t - \phi_{t+1}) + \sum_{t=1}^T \frac{\eta}{2\alpha_h} \|\nabla f_t(x_t)\|_*^2 \\
&= \phi_1 - \phi_{T+1} + \frac{\eta \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2}{2\alpha_h} \leq \phi_0 + \frac{\eta \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2}{2\alpha_h} \\
&= \frac{D_h(x^* \| x_0)}{\eta} + \frac{\eta \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2}{2\alpha_h}.
\end{aligned}$$

The last inequality above uses that the Bregman divergence is always non-negative for convex functions (so $\phi_{T+1} = \frac{D_h(x^* \| x_{T+1})}{\eta} \geq 0$), and that $\phi_1 = \phi_0$ by definition of the potential. Hence the proof. $\blacksquare$

Conclusion 4.7. When the norm $\|\cdot\|$ is the ℓ_2 -norm and $h(x) = \frac{1}{2} \|x\|_2^2$, the average regret is

$$\frac{\|x^* - x_0\|_2^2}{2\eta T} + \frac{\eta \sum_{t=1}^T \|\nabla f_t(x_t)\|_2^2}{2T},$$

which is what Theorem 2.1 guarantees.

Conclusion 4.8. When the norm $\|\cdot\|$ is the ℓ_1 -norm and $h(x) := \sum_{i=1}^n x_i \ln x_i$ is the negative entropy, the regret versus any point $x^* \in \Delta_n$ is

$$\frac{1}{\eta} \sum_{i=1}^n x_i^* \ln \frac{x_i^*}{(x_0)_i} + \frac{\eta \sum_{t=1}^T \|\nabla f_t(x_t)\|_\infty^2}{2/\ln 2}.$$

For linear functions $f_t(x) = \langle \ell_t, x \rangle$ with $\ell_t \in [-1, 1]^n$, and $x_0 = 1/n \cdot (1, \dots, 1)^T$, the regret is

$$\frac{KL(x^* \| x_0)}{\eta} + \frac{\eta T}{2/\ln 2} \leq \frac{\ln n}{\eta} + \eta T,$$

where $KL(x^* \| x_0)$ is the KL-divergence from x^* to x_0 . The inequality uses that the KL-divergence to the uniform distribution is at most $\ln n$. From this, a new way to improve the regret bound is obtained: if we start with a distribution x_0 that is closer to x^* , the first term of the regret gets smaller.

4.2 A Parenthesis: Smooth Functions and General Norms

Our goal now is to minimize a function that is smooth with respect to an arbitrary norm (i.e., not necessarily the Euclidean norm), in the unconstrained case. The definition of a smooth function (3.1) can be extended naturally, when we consider an arbitrary norm $\|\cdot\|$. The following update rule can be obtained by extending (2.2):

$$x_{t+1} \leftarrow \arg \min_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - x_t\|^2 + \eta_t \langle \nabla f(x_t), x - x_t \rangle \right\}, \quad (4.6)$$

where $\|\cdot\|$ is an arbitrary norm. To emulate the minimum on the right side, we can use basic Fenchel duality defined as follows.

Definition 4.9 (Fenchel Duality). Given a function g , its *Fenchel dual* is defined as

$$g^*(\theta) := \max\{\langle \theta, z \rangle - g(z)\}.$$

We show now the general-norm analog of Theorem 3.3.

Theorem 4.10 (GD: Smooth Functions for General Norms). *If f is a function that is β -smooth with respect to the norm $\|\cdot\|$, and $D := \max_{x \in \mathbb{R}^n} \{\|x - x^*\| \mid f(x) \leq f(x_0)\}$, the update rule (4.6) with a time-invariant multiplier $\eta_t = \frac{1}{\beta}$ ensures*

$$f(x_T) - f(x^*) \leq 2\beta \frac{D^2}{T+1}.$$

Proof. Consider the following potential

$$\boxed{\phi_t = t(t+1) \cdot (f(x_t) - f(x^*))}.$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (t+1)(t+2) \cdot (f(x_{t+1}) - f(x^*)) - t(t+1) \cdot (f(x_t) - f(x^*)) \\ &= (t+1)(t+2) \cdot (f(x_{t+1}) - f(x_t)) + 2(t+1) \cdot (f(x_t) - f(x^*)). \end{aligned} \quad (4.7)$$

To bound the first term, we consider the function $g(z) = \frac{1}{2} \|z\|^2$, and it is known that $g^*(\theta) = \frac{1}{2} \|\theta\|_*^2$ is its Fenchel dual. Hence,

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - x_t\|^2 + \eta_t \langle \nabla f(x_t), x - x_t \rangle \right\} \\ = - \max_{x \in \mathbb{R}^n} \left\{ \eta_t \langle \nabla f(x_t), x_t - x \rangle - \frac{1}{2} \|x_t - x\|^2 \right\} \\ \underbrace{z = x_t - x}_{=} - \max_{z \in \mathbb{R}^n} \left\{ \langle \eta_t \nabla f(x_t), z \rangle - \frac{1}{2} \|z\|^2 \right\} = -\frac{1}{2} \|\eta_t \nabla f(x_t)\|_*^2. \end{aligned} \quad (4.8)$$

Since f is a function that is β -smooth and by setting $\eta_t = \frac{1}{\beta}$, we have

$$\begin{aligned} f(x_{t+1}) &\leq f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 \\ &= f(x_t) + \beta \cdot \left(\langle \eta_t \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{1}{2} \|x_{t+1} - x_t\|^2 \right) \\ &= f(x_t) + \beta \cdot \left(-\frac{1}{2} \|\eta_t \nabla f(x_t)\|_*^2 \right), \end{aligned}$$

where the last equality uses that x_{t+1} is the minimizer of the expression in (4.8). Therefore, we get

$$f(x_{t+1}) - f(x_t) \leq -\frac{1}{2\beta} \|\nabla f(x_t)\|_*^2. \quad (4.9)$$

For the second term in (4.7), we use convexity and generalized Cauchy-Schwartz:

$$\begin{aligned} f(x_t) - f(x^*) &\leq \langle \nabla f(x_t), x_t - x^* \rangle = \langle x_t - x^*, \nabla f(x_t) \rangle \leq \overbrace{\|x_t - x^*\| \cdot \|\nabla f(x_t)\|_*}^{=a\|x_t - x^*\| \cdot 1/a \|\nabla f(x_t)\|_*} \\ &\leq 1/2(a\|x_t - x^*\|^2 + (1/a)\|\nabla f(x_t)\|_*^2), \end{aligned}$$

for any parameter $a > 0$. Note that (4.9) implies that $f(x_t) \leq f(x_{t-1}) \leq \dots \leq f(x_0)$, so let us define $D := \max_{x \in \mathbb{R}^n} \{\|x - x^*\| \mid f(x) \leq f(x_0)\}$. So the potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &\leq (t+1)(t+2) \cdot \left(-\frac{1}{2\beta}\right) \|\nabla f(x_t)\|_*^2 \\ &\quad + 2(t+1) \cdot \left(\frac{1}{2}(aD^2 + \|\nabla f(x_t)\|_*^2/a)\right) \\ &= (t+1)(t+2) \cdot \left(-\frac{1}{2\beta}\right) \|\nabla f(x_t)\|_*^2 + (t+1) \cdot (aD^2 + \|\nabla f(x_t)\|_*^2/a) \quad (4.10) \end{aligned}$$

Choosing $a = \frac{2\beta}{t+2}$ cancels the gradient terms. Hence,

$$\phi_{t+1} - \phi_t \leq 2\beta D^2 \frac{t+1}{t+2} \leq 2\beta D^2,$$

where the last inequality holds since $\frac{t+1}{t+2} \leq 1$ for all t . By summing over t and using $\phi_0 = 0$, we have

$$\phi_T = \phi_T - \phi_0 = \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 2\beta D^2 = 2\beta D^2 \cdot T.$$

Hence, by definition of the potential:

$$f(x_T) - f(x^*) = \frac{\phi_T}{T(T+1)} \leq \frac{2\beta D^2 \cdot T}{T(T+1)} = 2\beta \frac{D^2}{T+1}.$$

Hence the proof. ■

4.3 A Parenthesis: Well-Conditioned Functions and General Norms

Let us now consider a function that is well-conditioned with respect to an arbitrary norm (i.e., not necessarily the Euclidean norm), in the unconstrained

case. The definition of well-conditioned functions in Section 3.2 can be extended naturally, when we consider an arbitrary norm $\|\cdot\|$. The following update rule can be obtained by extending (2.2) (with a fixed step size $\eta_t = \frac{1}{\beta}$):

$$x_{t+1} \leftarrow \arg \min_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - x_t\|^2 + \frac{1}{\beta} \langle \nabla f(x_t), x - x_t \rangle \right\}, \quad (4.11)$$

where $\|\cdot\|$ is an arbitrary norm. To emulate the minimum on the right side, Fenchel duality is also used here. We show now the general-norm analog of Theorem 3.9.

Theorem 4.11 (GD: Well-Conditioned Functions for General Norms). *If f is both α -strongly convex and β -smooth with respect to the norm $\|\cdot\|$, define $\kappa := \beta/\alpha$. The update rule (4.11) ensures*

$$f(x_T) - f(x^*) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*)) \quad (4.12)$$

for all x^* .

Proof. We assume $\kappa > 1$ for simplicity, since $\kappa = 1 \Leftrightarrow f(x) = ax^T x + b^T x + c$ for suitable scalars a, c and $b \in \mathbb{R}^n$. In this case, it is easily checked that the optimum solution x^* is achieved in a single step. We define $\gamma = \frac{1}{\kappa-1}$ and use the potential

$$\phi_t = (1 + \gamma)^t \cdot (f(x_t) - f(x^*)). \quad (4.13)$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (1 + \gamma)^{t+1} \cdot (f(x_{t+1}) - f(x^*)) - (1 + \gamma)^t \cdot (f(x_t) - f(x^*)) \\ &= (1 + \gamma)^t ((1 + \gamma)(f(x_{t+1}) - f(x_t)) + \gamma(f(x_t) - f(x^*))). \end{aligned} \quad (4.14)$$

To bound the first term, we consider the function $g(z) = \frac{1}{2} \|z\|^2$, and it is known that $g^*(\theta) = \frac{1}{2} \|\theta\|_*^2$ is its Fenchel dual. Hence,

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \left\{ \frac{1}{2} \|x - x_t\|^2 + \frac{1}{\beta} \langle \nabla f(x_t), x - x_t \rangle \right\} \\ = - \max_{x \in \mathbb{R}^n} \left\{ \frac{1}{\beta} \langle \nabla f(x_t), x_t - x \rangle - \frac{1}{2} \|x_t - x\|^2 \right\} \\ \stackrel{z = x_t - x}{=} - \max_{z \in \mathbb{R}^n} \left\{ \left\langle \frac{1}{\beta} \nabla f(x_t), z \right\rangle - \frac{1}{2} \|z\|^2 \right\} = -\frac{1}{2} \left\| \frac{1}{\beta} \nabla f(x_t) \right\|_*^2. \end{aligned} \quad (4.15)$$

Since f is in particular β -smooth, we have

$$\begin{aligned} f(x_{t+1}) &\leq f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{\beta}{2} \|x_{t+1} - x_t\|^2 \\ &= f(x_t) + \beta \cdot \left(\left\langle \frac{1}{\beta} \nabla f(x_t), x_{t+1} - x_t \right\rangle + \frac{1}{2} \|x_{t+1} - x_t\|^2 \right) \\ &= f(x_t) + \beta \cdot \left(-\frac{1}{2} \left\| \frac{1}{\beta} \nabla f(x_t) \right\|_*^2 \right), \end{aligned}$$

where the last equality uses that x_{t+1} is the minimizer of the expression in (4.15). Therefore, we get

$$f(x_{t+1}) - f(x_t) \leq -\frac{1}{2\beta} \|\nabla f(x_t)\|_*^2. \quad (4.16)$$

To bound the second term, we use the strong convexity of f with $x = x_t$ and $y = x^*$:

$$\begin{aligned} f(x_t) - f(x^*) &\leq \langle \nabla f(x_t), x_t - x^* \rangle - \frac{\alpha}{2} \|x_t - x^*\|^2 \\ &= \langle x_t - x^*, \nabla f(x_t) \rangle - \frac{\alpha}{2} \|x_t - x^*\|^2 \leq \frac{1}{2\alpha} \|\nabla f(x_t)\|_*^2, \end{aligned}$$

where the second inequality uses $\langle a, b \rangle - \|a\|^2/2 \leq \|b\|_*^2/2$ with $a = \sqrt{\alpha}(x_t - x^*)$ and $b = \frac{1}{\sqrt{\alpha}} \nabla f(x_t)$. Plugging those bounds back to (4.14) gives:

$$\phi_{t+1} - \phi_t \leq (1 + \gamma)^t \cdot \left(-\frac{1 + \gamma}{2\beta} + \frac{\gamma}{2\alpha} \right) \|\nabla f(x_t)\|_*^2.$$

Since $\gamma = \frac{1}{\kappa-1} = \frac{\frac{\beta}{\alpha}-1}{\frac{\beta}{\alpha}}$, we have

$$\begin{aligned} -\frac{1 + \gamma}{2\beta} + \frac{\gamma}{2\alpha} &= \frac{-\alpha(1 + \gamma) + \beta\gamma}{2\beta\alpha} = \frac{-\alpha \left(1 + \frac{1}{\beta/\alpha-1}\right) + \frac{\beta}{\beta/\alpha-1}}{2\beta\alpha} \\ &= \frac{-\alpha + \frac{\beta-\alpha}{1/\alpha(\beta-\alpha)}}{2\beta\alpha} = \frac{-\alpha + \frac{1}{1/\alpha}}{2\beta\alpha} = \frac{\overbrace{-\alpha + \alpha}^{=0}}{2\beta\alpha} = 0 \end{aligned}$$

Thus, $\phi_{t+1} - \phi_t \leq 0$ for all t and after T steps

$$\begin{aligned} \phi_T - \phi_0 &= \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 0 = 0 \\ &\Rightarrow \phi_T \leq \phi_0 \end{aligned}$$

So by definition of ϕ_t , we have

$$\begin{aligned} (1 + \gamma)^T (f(x_T) - f(x^*)) &\leq (1 + \gamma)^0 (f(x_0) - f(x^*)) = f(x_0) - f(x^*) \\ &\Rightarrow f(x_T) - f(x^*) \leq (1 + \gamma)^{-T} (f(x_0) - f(x^*)) \\ &= \left(1 + \frac{1}{\kappa - 1}\right)^{-T} (f(x_0) - f(x^*)) = \left(\frac{\kappa - 1}{\kappa - 1} + \frac{1}{\kappa - 1}\right)^{-T} (f(x_0) - f(x^*)) \\ &= \left(\frac{\kappa}{\kappa - 1}\right)^{-T} (f(x_0) - f(x^*)) = \left(\frac{\kappa - 1}{\kappa}\right)^T (f(x_0) - f(x^*)) \\ &= \left(1 - \frac{1}{\kappa}\right)^T (f(x_0) - f(x^*)) \leq \exp(-T/\kappa) \cdot (f(x_0) - f(x^*)). \end{aligned}$$

Hence the proof. ■

5 Nesterov Acceleration: A Potential Function Proof

In Section 3, we mentioned that there is a convergence rate of $O(1/T)$ for smooth functions, using both projected gradient descent and the Frank-Wolfe method. However, the lower bound is only $\Omega(1/T^2)$. In this case, the algorithm can be improved, and Yuri Nesterov showed how to do so using his *accelerated gradient descent* methods [5]. It is important to note that acceleration for non-smooth functions is not possible in general. Table 1 describes convergence rates of gradient descent against accelerated GD for functions of different types as a function of the number of iterations regardless of other constants and parameters (such as the Lipschitz constant, the initial distance and the diameter of constrained set).

	general convex	α -strongly convex	β -smooth	κ -well conditioned
Gradient descent	$\frac{1}{\sqrt{T}}$	$\frac{1}{\alpha T}$	$\frac{\beta}{T}$	$\exp(-T/\kappa)$
Accelerated GD	—	—	$\frac{\beta}{T^2}$	$\exp(-T/\sqrt{\kappa})$

Table 1: Rates of Convergence of Gradient Descent Against Accelerated GD.

We show now potential-based proofs for his theorem, both for the smooth case, and for the well-conditioned case. At first we consider only the Euclidean norm, and later we extend to general norms in the unconstrained case.

5.1 Accelerated Gradient Descent

The main idea of the accelerated gradient descent is to run two iterative processes for y_t and z_t , and then to combine them together to get the concrete point x_t . We start with $x_0 = y_0 = z_0$, and at timestep t , we use x_t . Now, the update rules are defined as follows:

$$y_{t+1} \leftarrow x_t - \frac{1}{\beta} \nabla f(x_t), \quad (5.1)$$

$$z_{t+1} \leftarrow z_t - \eta_t \nabla f(x_t), \quad (5.2)$$

$$x_{t+1} \leftarrow (1 - \tau_{t+1})y_{t+1} + \tau_{t+1}z_{t+1}. \quad (5.3)$$

In (5.2), we will use an “aggressive” step size $\eta_t = \frac{t+1}{2\beta}$. In (5.3), the mixing weight is $\tau_t = \frac{2}{t+2}$. The following convergence bound can be obtained using those update rules for smooth functions.

Theorem 5.1 (Accelerated GD). *Given a β -smooth function f , the update rules (5.1)-(5.3) ensure*

$$f(y_T) - f(x^*) \leq 2\beta \frac{\|z_0 - x^*\|^2}{T(T+1)}.$$

Proof. Consider the following potential

$$\boxed{\phi_t = t(t+1) \cdot (f(y_t) - f(x^*)) + 2\beta \cdot \|z_t - x^*\|^2}. \quad (5.4)$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (t+1)(t+2) \cdot (f(y_{t+1}) - f(x^*)) + 2\beta \cdot \|z_{t+1} - x^*\|^2 \\ &\quad - t(t+1) \cdot (f(y_t) - f(x^*)) - 2\beta \|z_t - x^*\|^2 = t(t+1) \cdot (f(y_{t+1}) - f(y_t)) \\ &\quad + 2(t+1) \cdot (f(y_{t+1}) - f(x^*)) + 2\beta (\|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2) \end{aligned} \quad (5.5)$$

Using Fact 1.1 with $a = -\eta_t \nabla f(x_t)/\sqrt{2}$ and $b = (z_t - x^*)/\sqrt{2}$ ($a + b = (z_t - \eta_t \nabla f(x_t) - x^*)/\sqrt{2} = (z_{t+1} - x^*)/\sqrt{2}$), we have

$$\begin{aligned} \frac{1}{2} \|z_{t+1} - x^*\|^2 &= \frac{\eta_t^2}{2} \|\nabla f(x_t)\|^2 + \langle -\eta_t \nabla f(x_t), z_t - x^* \rangle + \frac{1}{2} \|z_t - x^*\|^2 \\ \Rightarrow \frac{1}{2} (\|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2) &= \frac{\eta_t^2}{2} \|\nabla f(x_t)\|^2 + \eta_t \langle \nabla f(x_t), x^* - z_t \rangle. \end{aligned} \quad (5.6)$$

Plugging this back to (5.5) gives:

$$\begin{aligned} \phi_{t+1} - \phi_t &= t(t+1) \cdot (f(y_{t+1}) - f(y_t)) + 2(t+1) \cdot (f(y_{t+1}) - f(x^*)) \\ &\quad + 4\beta \left(\frac{\eta_t^2}{2} \|\nabla f(x_t)\|^2 + \eta_t \langle \nabla f(x_t), x^* - z_t \rangle \right). \end{aligned}$$

To bound the first term and the second term, we use the smoothness of f with $x = x_t$ and $y = y_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$:

$$\begin{aligned} f(y_{t+1}) &\leq f(x_t) + \left\langle \nabla f(x_t), \underbrace{y_{t+1} - x_t}_{= -\nabla f(x_t)/\beta} \right\rangle + \frac{\beta}{2} \left\| \underbrace{y_{t+1} - x_t}_{= -\nabla f(x_t)/\beta} \right\|^2 \\ &= f(x_t) - \frac{1}{\beta} \|\nabla f(x_t)\|^2 + \frac{1}{2\beta} \|\nabla f(x_t)\|^2 = f(x_t) - \frac{1}{2\beta} \|\nabla f(x_t)\|^2. \end{aligned}$$

So the potential change is

$$\begin{aligned}
\phi_{t+1} - \phi_t &\leq t(t+1) \left(f(x_t) - \frac{1}{2\beta} \|\nabla f(x_t)\|^2 - f(y_t) \right) \\
&\quad + 2(t+1) \left(f(x_t) - \frac{1}{2\beta} \|\nabla f(x_t)\|^2 - f(x^*) \right) + 2\beta\eta_t^2 \|\nabla f(x_t)\|^2 \\
&\quad + 4\beta\eta_t \langle \nabla f(x_t), x^* - z_t \rangle \leq t(t+1)(f(x_t) - f(y_t)) + 2(t+1)(f(x_t) - f(x^*)) \\
&\quad - \frac{(t+1)(t+2)}{2\beta} \|\nabla f(x_t)\|^2 + \frac{(t+1)(t+2)}{2\beta} \|\nabla f(x_t)\|^2 + 2(t+1) \cdot \langle \nabla f(x_t), x^* - z_t \rangle \\
&\leq t(t+1) \cdot \langle \nabla f(x_t), x_t - y_t \rangle + 2(t+2) \cdot \langle \nabla f(x_t), x_t - x^* \rangle + 2(t+1) \cdot \langle \nabla f(x_t), x^* - z_t \rangle.
\end{aligned}$$

The second inequality uses that $2\beta\eta_t^2 = \frac{(t+1)^2}{2\beta} \leq \frac{(t+1)(t+2)}{2\beta}$, and the last inequality uses convexity for the first two terms. By collecting like terms, we get

$$\begin{aligned}
\phi_{t+1} - \phi_t &\leq (t+1) \cdot \langle \nabla f(x_t), tx_t - ty_t + 2x_t - 2x^* + 2x^* - 2z_t \rangle \\
&= (t+1) \cdot \left\langle \nabla f(x_t), \underbrace{(t+2)x_t - ty_t - 2z_t}_{=0} \right\rangle = 0,
\end{aligned}$$

using (5.3) and $\tau_t = \frac{2}{t+2}$. So, $\phi_{t+1} - \phi_t \leq 0$ for all t . By summing over t and using $\phi_0 = 2\beta \cdot \|z_0 - x^*\|^2$, we have

$$\begin{aligned}
\phi_T - 2\beta \cdot \|z_0 - x^*\|^2 &= \phi_T - \phi_0 = \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 0 = 0 \\
&\Rightarrow \phi_T \leq 2\beta \cdot \|z_0 - x^*\|^2.
\end{aligned}$$

Hence, by definition of the potential:

$$\begin{aligned}
f(y_T) - f(x^*) &= \frac{\phi_T}{T(T+1)} - 2\beta \cdot \frac{\|z_T - x^*\|^2}{T(T+1)} \\
&\quad \text{by removing the negative} \\
&\quad \text{squard-norm term} \\
&\leq \frac{\phi_T}{T(T+1)} \leq 2\beta \cdot \frac{\|z_0 - x^*\|^2}{T(T+1)}.
\end{aligned}$$

Hence the proof. ■

5.1.1 A Parenthesis: Optimizing Parameters and Making Connections

Consider the following generic potential

$$\phi_t = \lambda_{t-1}^2 (f(y_t) - f(x^*)) + \frac{\beta}{2} \|z_t - x^*\|^2,$$

where $\lambda_t = \Theta(t^2)$, and the goal is to optimize the calculation above. Having $\lambda_t^2 - \lambda_{t-1}^2 = \lambda_t$ (which is equivalent to $\lambda_t^2 - \lambda_t - \lambda_{t-1}^2 = 0$) and $\tau_t = \frac{1}{\lambda_t}$ makes the calculations easy to calculate. Solving this recurrence gives the following weights

$$\lambda_0 = 0, \quad \lambda_t = \frac{1 + \sqrt{1 + 4\lambda_{t-1}^2}}{2} \quad (5.7)$$

In the literature the update rules (5.1)-(5.3) have sometimes been called *AGM2* (Neterov's second accelerated method), and these weights are used in the standard exposition of them. Let's define a different set of update rules named *AGM1* for the optimize choice of λ_t from (5.7):

$$\boxed{\begin{aligned} y_{t+1} &\leftarrow x_t - \frac{1}{\beta} \nabla f(x_t), \\ x_{t+1} &\leftarrow \left(1 - \frac{1 - \lambda_t}{\lambda_{t+1}}\right) y_{t+1} + \frac{1 - \lambda_t}{\lambda_{t+1}} y_t. \end{aligned}} \quad (5.8)$$

$$(5.9)$$

There is an obvious equivalence between these two sets of update rules.

Lemma 5.2. *Using the update rules (5.8)-(5.9) and setting $z_t := \lambda_t x_t - (\lambda_t - 1)y_t = \lambda_t(x_t - y_t) + y_t$, $\tau_t := \frac{1}{\lambda_t}$ and $\eta_t := \frac{1}{\beta\tau_t}$ leads to the update rules (5.1)-(5.3).*

Proof. The update rule for y_t is identical, so it sufficient to show that the z_t and x_t behave in the same manner. According to the definition of z_t and since $\tau_t = \frac{1}{\lambda_t}$, we have $\tau_t z_t = x_t - y_t + \tau_t y_t$. So,

$$x_t = \frac{1}{\tau_t} z_t + (1 - \tau_t) y_t, \quad \text{and}$$

$$z_{t+1} - z_t = (\lambda_{t+1}(x_{t+1} - y_{t+1}) + y_{t+1}) - (\lambda_t(x_t - y_t) + y_t). \quad (5.10)$$

Also, multiplying (5.9) by λ_{t+1} gives

$$\lambda_{t+1} x_{t+1} = (\lambda_{t+1} - 1 + \lambda_t) y_{t+1} + (1 - \lambda_t) y_t$$

and thus

$$\lambda_{t+1}(x_{t+1} - y_{t+1}) - (1 - \lambda_t)(y_t - y_{t+1}) = 0. \quad (5.11)$$

Subtracting (5.11) from (5.10) gives

$$z_{t+1} - z_t = (z_{t+1} - z_t) - 0 = \lambda_t y_{t+1} - \lambda_t x_t = \lambda_t (y_{t+1} - x_t) \stackrel{(5.8)}{=} -\frac{\lambda_t}{\beta} \nabla f(x_t).$$

Since $\lambda_t = \frac{1}{\tau_t} = \beta \eta_t$, this is exactly the update rule $z_{t+1} \leftarrow z_t - \eta_t \nabla f(x_t)$. So there is equivalence for both update rules. $\blacksquare$

5.2 The Constrained Case With Acceleration

Given a convex set $K \subseteq \mathbb{R}^n$, the update rules are similar to the update rules for the unconstrained case, they just include projecting the points onto the set K . Again, we start with $x_0 = y_0 = z_0$, and at timestep t , we use x_t . Now, the update rules are defined as follows:

$$y_{t+1} \leftarrow \Pi_K(x_t - \frac{1}{\beta} \nabla f(x_t)), \quad (5.12)$$

$$z_{t+1} \leftarrow \Pi_K(z_t - \eta_t \nabla f(x_t)), \quad (5.13)$$

$$x_{t+1} \leftarrow (1 - \tau_{t+1})y_{t+1} + \tau_{t+1}z_{t+1}. \quad (5.14)$$

Here we use the same “aggressive” step size $\eta_t = \frac{t+1}{2\beta}$ and the same mixing weight $\tau_t = \frac{2}{t+2}$. Using these update rules we can obtain the same convergence bound as in Theorem 5.1 for the unconstrained case.

Theorem 5.3 (Accelerated GD: Constrained Optimization). *Given a function $f : K \rightarrow \mathbb{R}$ over a convex set $K \subseteq \mathbb{R}^n$ that is β -smooth, the update rules (5.12)-(5.14) ensure*

$$f(y_T) - f(x^*) \leq 2\beta \cdot \frac{\|z_0 - x^*\|^2}{T(T+1)}.$$

Proof. Consider the following potential

$$\phi_t = t(t+1) \cdot (f(y_t) - f(x^*)) + 2\beta \cdot \|z_t - x^*\|^2. \quad (5.15)$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (t+1)(t+2) \cdot (f(y_{t+1}) - f(x^*)) + 2\beta \cdot \|z_{t+1} - x^*\|^2 \\ &\quad - t(t+1) \cdot (f(y_t) - f(x^*)) - 2\beta \cdot \|z_t - x^*\|^2 = \underbrace{(t+1)(t+2) \cdot (f(y_{t+1}) - f(x_t))}_{(\star)} \\ &\quad - \underbrace{t(t+1) \cdot (f(y_t) - f(x_t)) + 2(t+1) \cdot (f(x_t) - f(x^*))}_{(\star\star)} + \underbrace{2\beta \cdot (\|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2)}_{(\star\star\star)}. \end{aligned} \quad (5.16)$$

Using convexity on both terms in $(\star\star)$ gives

$$\begin{aligned} (\star\star) &\leq -t(t+1) \cdot \langle \nabla f(x_t), y_t - x_t \rangle + 2(t+1) \cdot \langle \nabla f(x_t), x_t - x^* \rangle \\ &= (t+1) \cdot \langle \nabla f(x_t), -t(y_t - x_t) + 2(x_t - x^*) \rangle. \end{aligned}$$

²Originally in [1] it was stated that setting $\eta_t = \frac{t+1}{\beta}$ with the same mixing weight gives the result. However, I believe this is an error, because I tried to prove this theorem myself using the same potential function and I only achieved the result by setting $\eta_t = \frac{t+1}{2\beta}$.

Now by (5.14) $(1 - \tau_t)(y_t - x_t) = \tau_t(x_t - z_t)$. If $\tau_t = \frac{2}{t+2}$, then we have $t(y_t - x_t) = 2(x_t - z_t)$, and thus

$$(\star\star) \leq (t+1) \cdot \langle \nabla f(x_t), -2(x_t - z_t) + 2(x_t - x^*) \rangle = 2(t+1) \cdot \langle \nabla f(x_t), z_t - x^* \rangle.$$

Using Fact 1.1 with $a = z_t - z_{t+1}$ and $b = z_{t+1} - x^*$ ($a + b = z_t - x^*$) gives

$$\begin{aligned} \|z_t - x^*\|^2 &= \|z_t - z_{t+1}\|^2 + 2 \langle z_t - z_{t+1}, z_{t+1} - x^* \rangle + \|z_{t+1} - x^*\|^2 \\ &\Rightarrow \|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2 = 2 \langle z_{t+1} - z_t, z_{t+1} - x^* \rangle - \|z_{t+1} - z_t\|^2. \end{aligned}$$

Therefore, the expression $(\star\star\star)$ is

$$2\beta \cdot \left(2 \langle z_{t+1} - z_t, z_{t+1} - x^* \rangle - \|z_{t+1} - z_t\|^2 \right). \quad (5.17)$$

By Proposition 1.6, $\langle \tilde{z}_{t+1} - z_{t+1}, z_{t+1} - x^* \rangle \geq 0$ where $\tilde{z}_{t+1} = z_t - \eta_t \nabla f(x_t)$. Multiplying this by 4β and adding to (5.17), and setting $\eta_t = \frac{t+1}{2\beta}$, we get

$$\begin{aligned} (\star\star\star) &\leq 4\beta \cdot \langle z_{t+1} - z_t, z_{t+1} - x^* \rangle - 2\beta \cdot \|z_{t+1} - z_t\|^2 + 4\beta \cdot \langle \tilde{z}_{t+1} - z_{t+1}, z_{t+1} - x^* \rangle \\ &= 4\beta \cdot \left\langle z_{t+1} - z_t + \underbrace{z_t - \eta_t \nabla f(x_t)}_{=\tilde{z}_{t+1}} - z_{t+1}, z_{t+1} - x^* \right\rangle - 2\beta \cdot \|z_{t+1} - z_t\|^2 \\ &= -2(t+1) \cdot \langle \nabla f(x_t), z_{t+1} - x^* \rangle - 2\beta \cdot \|z_{t+1} - z_t\|^2. \end{aligned}$$

Plugging the bounds of $(\star\star)$ and $(\star\star\star)$ back to (5.16) gives:

$$\begin{aligned} \phi_{t+1} - \phi_t &\leq \underbrace{(t+1)(t+2) \cdot (f(y_{t+1}) - f(x_t))}_{(\star)} + \underbrace{2(t+1) \cdot \langle \nabla f(x_t), z_t - x^* \rangle}_{(\star\star)} \\ &\quad \underbrace{- 2(t+1) \cdot \langle \nabla f(x_t), z_{t+1} - x^* \rangle - 2\beta \cdot \|z_{t+1} - z_t\|^2}_{(\star\star\star)} = \\ &\quad (t+1)(t+2) \cdot (f(y_{t+1}) - f(x_t)) + 2(t+1) \cdot \langle \nabla f(x_t), z_t - z_{t+1} \rangle \\ &\quad - 2\beta \cdot \|z_{t+1} - z_t\|^2 \end{aligned}$$

So to show that $\phi_{t+1} - \phi_t \leq 0$, it suffices to show that

$$(t+1)(t+2) \cdot (f(y_{t+1}) - f(x_t)) + 2(t+1) \cdot \langle \nabla f(x_t), z_t - z_{t+1} \rangle - 2\beta \cdot \|z_{t+1} - z_t\|^2 \leq 0$$

To do so, we use the smoothness of f with $y = y_{t+1}$ and $x = x_t$:

$$f(y_{t+1}) - f(x_t) \leq \langle \nabla f(x_t), y_{t+1} - x_t \rangle + \frac{\beta}{2} \|y_{t+1} - x_t\|^2.$$

Since

$$\min_{y \in K} \left\{ \langle \nabla f(x_t), y - x_t \rangle + \frac{\beta}{2} \|y - x_t\|^2 \right\}$$

is minimized by the definition of y_{t+1} , the following inequality holds for any $v \in K$,

$$\langle \nabla f(x_t), y_{t+1} - x_t \rangle + \frac{\beta}{2} \|y_{t+1} - x_t\|^2 \leq \langle \nabla f(x_t), v - x_t \rangle + \frac{\beta}{2} \|v - x_t\|^2$$

and hence

$$f(y_{t+1}) - f(x_t) \leq \langle \nabla f(x_t), v - x_t \rangle + \frac{\beta}{2} \|v - x_t\|^2.$$

Consider $v := (1 - \tau_t)y_t + \tau_t z_{t+1} \in K$, so $v - x_t = \tau_t(z_{t+1} - z_t)$. Substituting, we get

$$f(y_{t+1}) - f(x_t) \leq \tau_t \langle \nabla f(x_t), z_{t+1} - z_t \rangle + \frac{\tau_t^2 \beta}{2} \|z_{t+1} - z_t\|^2.$$

Setting $\tau_t = \frac{2}{t+2}$, we get

$$\begin{aligned} f(y_{t+1}) - f(x_t) &\leq \frac{2}{t+2} \langle \nabla f(x_t), z_{t+1} - z_t \rangle + \frac{4\beta}{2(t+2)^2} \|z_{t+1} - z_t\|^2 \\ &= \frac{2}{t+2} \langle \nabla f(x_t), z_{t+1} - z_t \rangle + \frac{2\beta}{(t+2)^2} \|z_{t+1} - z_t\|^2. \end{aligned}$$

Multiplying by $(t+1)(t+2)$ and moving the expression in the RHS to the LHS give

$$\begin{aligned} (t+1)(t+2) \cdot (f(y_{t+1}) - f(x_t)) &+ 2(t+1) \cdot \langle \nabla f(x_t), z_t - z_{t+1} \rangle \\ &+ 2\beta \frac{t+1}{t+2} \cdot \|z_{t+1} - z_t\|^2 \leq 0. \end{aligned}$$

Since $-2\beta \cdot \|z_{t+1} - z_t\|^2 \leq 2\beta \frac{t+1}{t+2} \cdot \|z_{t+1} - z_t\|^2$ for all t , we get

$$\begin{aligned} \phi_{t+1} - \phi_t &\leq (t+1)(t+2) \cdot (f(y_{t+1}) - f(x_t)) + 2(t+1) \cdot \langle \nabla f(x_t), z_t - z_{t+1} \rangle \\ &\quad - 2\beta \cdot \|z_{t+1} - z_t\|^2 \leq 0. \end{aligned}$$

So, $\phi_{t+1} - \phi_t \leq 0$ for all t . By summing over t and using $\phi_0 = 2\beta \cdot \|z_0 - x^*\|^2$, we have

$$\begin{aligned} \phi_T - 2\beta \cdot \|z_0 - x^*\|^2 &= \phi_T - \phi_0 = \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 0 = 0 \\ &\Rightarrow \phi_T \leq 2\beta \cdot \|z_0 - x^*\|^2. \end{aligned}$$

Hence, by definition of the potential:

$$\begin{aligned} f(y_T) - f(x^*) &= \frac{\phi_T}{T(T+1)} - 2\beta \cdot \frac{\|z_T - x^*\|^2}{T(T+1)} \\ &\quad \text{by removing the negative} \\ &\quad \text{squard-norm term} \\ &\leq \frac{\phi_T}{T(T+1)} \leq 2\beta \cdot \frac{\|z_0 - x^*\|^2}{T(T+1)}. \end{aligned}$$

Hence the proof. $\blacksquare$

5.3 The Extension to Arbitrary Norms

Given an arbitrary norm $\|\cdot\|$, the update rules now use the gradient descent update (4.6) for smooth functions for the y variables, and the mirror descent update rule (4.2) for the strongly-convex function h (where $f_t = f$ at each time t) for the z variables:

$$y_{t+1} \leftarrow \arg \min_{y \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y - x_t\|^2 + \frac{1}{\beta} \langle \nabla f(x_t), y - x_t \rangle \right\}, \quad (5.18)$$

$$z_{t+1} \leftarrow \arg \min_{z \in \mathbb{R}^n} \{ \langle \eta_t \nabla f(x_t), z - z_t \rangle + D_h(z \| z_t) \}, \quad (5.19)$$

$$x_{t+1} \leftarrow (1 - \tau_{t+1})y_{t+1} + \tau_{t+1}z_{t+1}. \quad (5.20)$$

Note that (5.19) is equivalent to the following update rules (where $\theta_t = \nabla h(z_t)$):

$$\begin{aligned} \tilde{\theta}_{t+1} &\leftarrow \theta_t - \eta_t \nabla f(x_t), \\ \tilde{z}_{t+1} &\leftarrow \nabla h^*(\tilde{\theta}_{t+1}), \\ z_{t+1} &\leftarrow \arg \min_{z \in \mathbb{R}^n} D_h(z \| \tilde{z}_{t+1}). \end{aligned} \quad (5.21)$$

Note that we consider the unconstrained case, so $\min_{z \in \mathbb{R}^n} D_h(z \| \tilde{z}_{t+1}) = D_h(\tilde{z}_{t+1} \| \tilde{z}_{t+1}) = 0$ and thus $z_{t+1} = \tilde{z}_{t+1}$. Here we use a different step size and the same mixing weight $\tau_t = \frac{2}{t+2}$. The following convergence bound can be achieved by setting the step size to

$$\eta_t = \frac{(t+1)\alpha_h}{2\beta},$$

where α_h is such that h is α_h -strongly convex with respect to $\|\cdot\|$.

Theorem 5.4 (Accelerated GD: General Norms). *Given a β -smooth function f with respect to norm $\|\cdot\|$, the update rules (5.18)-(5.20) ensure*

$$f(y_T) - f(x^*) \leq \frac{4\beta}{\alpha_h} \cdot \frac{D_h(x^* \| z_0) - D_h(x^* \| z_T)}{T(T+1)}.$$

Proof. Consider the following potential

$$\phi_t = t(t+1) \cdot (f(y_t) - f(x^*)) + \frac{4\beta}{\alpha_h} \cdot D_h(x^* \| z_t). \quad (5.22)$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (t+1)(t+2) \cdot (f(y_{t+1}) - f(x^*)) + \frac{4\beta}{\alpha_h} \cdot D_h(x^* \| z_{t+1}) \\ &\quad - t(t+1) \cdot (f(y_t) - f(x^*)) - \frac{4\beta}{\alpha_h} \cdot D_h(x^* \| z_t) = t(t+1) \cdot (f(y_{t+1}) - f(y_t)) \\ &\quad + 2(t+1) \cdot (f(y_{t+1}) - f(x^*)) + \frac{4\beta}{\alpha_h} \cdot (D_h(x^* \| z_{t+1}) - D_h(x^* \| z_t)). \end{aligned} \quad (5.23)$$

To bound the first term and the second term in (5.23), we consider the function $g(w) = \frac{1}{2} \|w\|^2$ and it is known that $g^*(\theta) = \frac{1}{2} \|\theta\|_*^2$ is its Fenchel dual. Hence,

$$\begin{aligned} & \min_{y \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y - x_t\|^2 + \frac{1}{\beta} \langle \nabla f(x_t), y - x_t \rangle \right\} \\ &= - \max_{y \in \mathbb{R}^n} \left\{ \frac{1}{\beta} \langle \nabla f(x_t), x_t - y \rangle - \frac{1}{2} \|x_t - y\|^2 \right\} \\ & \underbrace{w = x_t - y}_{=} - \max_{w \in \mathbb{R}^n} \left\{ \left\langle \frac{1}{\beta} \nabla f(x_t), w \right\rangle - \frac{1}{2} \|w\|^2 \right\} = -\frac{1}{2} \left\| \frac{1}{\beta} \nabla f(x_t) \right\|_*^2. \quad (5.24) \end{aligned}$$

Since f is a function that is β -smooth, we have

$$\begin{aligned} f(y_{t+1}) &\leq f(x_t) + \langle \nabla f(x_t), y_{t+1} - x_t \rangle + \frac{\beta}{2} \|y_{t+1} - x_t\|^2 \\ &= f(x_t) + \beta \left(\left\langle \frac{1}{\beta} \nabla f(x_t), y_{t+1} - x_t \right\rangle + \frac{1}{2} \|y_{t+1} - x_t\|^2 \right) \\ &= f(x_t) + \beta \cdot \left(-\frac{1}{2} \left\| \frac{1}{\beta} \nabla f(x_t) \right\|_*^2 \right), \end{aligned}$$

where the last equality uses that y_{t+1} is the minimizer of the expression in (5.24). Therefore, we get

$$f(y_{t+1}) \leq f(x_t) - \frac{1}{2\beta} \cdot \|\nabla f(x_t)\|_*^2. \quad (5.25)$$

For the third term in (5.23), we use the update rules (5.21) for the z variables

(instead of the x variables):

$$\begin{aligned}
& D_h(x^* \| z_{t+1}) - D_h(x^* \| z_t) \\
& \quad \underbrace{\text{by generalized}}_{\text{Pythagorean Property}} \leq D_h(x^* \| \tilde{z}_{t+1}) - D_h(x^* \| z_t) \\
& = h(x^*) - h(\tilde{z}_{t+1}) - \left\langle \underbrace{\nabla h(\tilde{z}_{t+1})}_{=\tilde{\theta}_{t+1}}, x^* - \tilde{z}_{t+1} \right\rangle - h(x^*) + h(z_t) \\
& \quad + \left\langle \underbrace{\nabla h(z_t)}_{=\theta_t}, x^* - z_t \right\rangle - \langle \nabla h(\tilde{z}_{t+1}), z_t - x^* \rangle + \underbrace{\langle \nabla h(\tilde{z}_{t+1}), z_t - x^* \rangle}_{=-\langle \tilde{\theta}_{t+1} x^* - z_t \rangle} \\
& = h(z_t) - h(\tilde{z}_{t+1}) - \left\langle \underbrace{\tilde{\theta}_{t+1}}_{=\theta_t - \eta_t \nabla f(z_t)}, z_t - \tilde{z}_{t+1} \right\rangle - \left\langle \underbrace{\tilde{\theta}_{t+1} - \theta_t}_{=-\eta_t \nabla f(z_t)}, x^* - z_t \right\rangle \\
& \quad \underbrace{h(z_t) - h(\tilde{z}_{t+1}) - \langle \theta_t, z_t - \tilde{z}_{t+1} \rangle}_{\leq (-\alpha_h/2) \|z_t - \tilde{z}_{t+1}\|^2 \text{ by strong convexity of } h} + \langle \eta_t \nabla f(z_t), z_t - \tilde{z}_{t+1} \rangle + \langle \eta_t \nabla f(z_t), x^* - z_t \rangle \\
& \leq -\frac{\alpha_h}{2} \|z_t - \tilde{z}_{t+1}\|^2 + \eta_t \langle \nabla f(z_t), z_t - \tilde{z}_{t+1} \rangle + \eta_t \langle \nabla f(z_t), x^* - z_t \rangle \\
& = -\frac{\alpha_h}{2} \|z_t - \tilde{z}_{t+1}\|^2 + \eta_t \langle z_t - \tilde{z}_{t+1}, \nabla f(z_t) \rangle + \eta_t \langle \nabla f(z_t), x^* - z_t \rangle \\
& \leq \frac{\eta_t^2}{2\alpha_h} \|\nabla f(z_t)\|_*^2 + \eta_t \langle \nabla f(z_t), x^* - z_t \rangle.
\end{aligned}$$

The last inequality uses generalized Cauchy-Schwartz to get $\langle a, b \rangle \leq \|a\| \cdot \|b\|_* \leq \|a\|^2/2 + \|b\|_*^2/2$ with $a = \sqrt{\alpha_h}(z_t - \tilde{z}_{t+1})$ and $b = \frac{\nabla f(z_t)}{\sqrt{\alpha_h}}$. So the potential change

is

$$\begin{aligned}
\phi_{t+1} - \phi_t &\leq t(t+1) \cdot \left(f(x_t) - \frac{1}{2\beta} \cdot \|\nabla f(x_t)\|_*^2 - f(y_t) \right) \\
&\quad + 2(t+1) \cdot \left(f(x_t) - \frac{1}{2\beta} \cdot \|\nabla f(x_t)\|_*^2 - f(x^*) \right) \\
&\quad + \frac{4\beta}{\alpha_h} \cdot \left(\frac{\eta_t^2}{2\alpha_h} \|\nabla f(x_t)\|_*^2 + \eta_t \langle \nabla f(x_t), \nabla f(z_t), x^* - z_t \rangle \right) \\
&\quad \underbrace{\text{by convexity of } f}_{\leq} \quad t(t+1) \cdot \langle \nabla f(x_t), x_t - y_t \rangle \\
&\quad + 2(t+1) \langle \nabla f(x_t), x_t - x^* \rangle - \underbrace{\frac{(t+2)(t+1)}{2\beta} \cdot \|\nabla f(x_t)\|_*^2}_{\text{}} \\
&\quad + \frac{2\beta\eta_t^2}{\alpha_h^2} \|\nabla f(x_t)\|_*^2 + \frac{4\beta\eta_t}{\alpha_h} \langle \nabla f(x_t), \nabla f(z_t), x^* - z_t \rangle \\
&\leq t(t+1) \cdot \langle \nabla f(x_t), x_t - y_t \rangle + 2(t+1) \langle \nabla f(x_t), x_t - x^* \rangle - \underbrace{\frac{(t+1)^2}{2\beta} \cdot \|\nabla f(x_t)\|_*^2}_{\text{}} \\
&\quad + \frac{(t+1)^2}{2\beta} \|\nabla f(x_t)\|_*^2 + 2(t+1) \langle \nabla f(x_t), \nabla f(z_t), x^* - z_t \rangle = \\
&\quad (t+1) \cdot \langle \nabla f(x_t), tx_t - ty_t + 2x_t - 2x^* + 2x^* - 2z_t \rangle \\
&\quad = (t+1) \cdot \left\langle \nabla f(x_t), \underbrace{(t+2)x_t - ty_t - 2z_t}_{=\vec{0}} \right\rangle = 0.
\end{aligned}$$

The last inequality uses that $\eta_t = \frac{(t+1)\alpha_h}{2\beta}$ and that $-\frac{(t+2)(t+1)}{2\beta} \cdot \|\nabla f(x_t)\|_*^2 \leq -\frac{(t+1)^2}{2\beta} \cdot \|\nabla f(x_t)\|_*^2$ for all t , and the last equality uses (5.20) and $\tau_t = \frac{2}{t+2}$. So, $\phi_{t+1} - \phi_t \leq 0$ for all t . By summing over t and using $\phi_0 = \frac{4\beta}{\alpha_h} \cdot D_h(x^* \| z_0)$, we have

$$\begin{aligned}
\phi_T - \frac{4\beta}{\alpha_h} \cdot D_h(x^* \| z_0) &= \phi_T - \phi_0 = \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 0 = 0 \\
&\Rightarrow \phi_T \leq \frac{4\beta}{\alpha_h} \cdot D_h(x^* \| z_0).
\end{aligned}$$

Hence, by definition of the potential:

$$f(y_T) - f(x^*) = \frac{\phi_T}{T(T+1)} - \frac{4\beta}{\alpha_h} \cdot \frac{D_h(x^* \| z_T)}{T(T+1)} \leq \frac{4\beta}{\alpha_h} \cdot \frac{D_h(x^* \| z_0) - D_h(x^* \| z_T)}{T(T+1)}.$$

Hence the proof. $\blacksquare$

5.4 Acceleration for Strongly Convex Functions

Now consider the case for which the function f is well-conditioned with conditioned number $\kappa = \frac{\beta}{\alpha}$. Our goal is to describe the algorithm of Nesterov with convergence rate $\exp(-T/\sqrt{\kappa})$ [6], where T is the number of iterations.

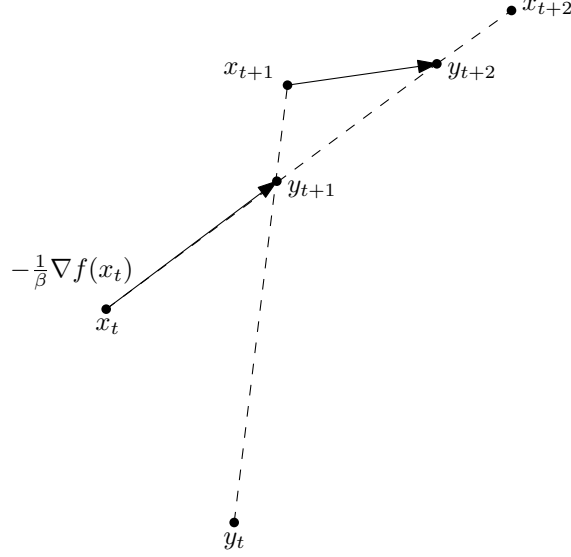


Figure 5: Nesterov's Accelerated Gradient Descent.

As mentioned above, this is the best possible for gradient methods. We consider the unconstrained case. The update rules are of the form of *AGM1* updates. So, the update rules are defined as follows:

$$y_{t+1} \leftarrow x_t - \frac{1}{\beta} \nabla f(x_t), \quad (5.26)$$

$$x_{t+1} \leftarrow \left(1 + \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right) y_{t+1} - \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} y_t. \quad (5.27)$$

This accelerated gradient method of Nesterov is illustrated in Figure 5. For convenience, we define $\tau = \frac{1}{\sqrt{\kappa} + 1}$ and $\gamma = \frac{1}{\sqrt{\kappa} - 1}$ for the analysis and set

$$z_{t+1} := \frac{1}{\tau} x_{t+1} - \frac{1 - \tau}{\tau} y_{t+1}. \quad (5.28)$$

Again, we start with $x_0 = y_0 = z_0$, and at timestep t , we use x_t . To show this best convergence guarantee, the following claim is used:

Claim 5.5. $z_{t+1} = (1 - \frac{1}{\sqrt{\kappa}})z_t + \frac{1}{\sqrt{\kappa}}x_t - \frac{1}{\alpha\sqrt{\kappa}}\nabla f(x_t)$ and so
 $z_{t+1} - x^* = \frac{1}{1+\gamma}(z_t - x^*) + \frac{\gamma}{1+\gamma}(x_t - x^*) - \frac{\gamma}{\alpha(1+\gamma)}\nabla f(x_t)$.

Proof. The expression of x_{t+1} from (5.27) is

$$\begin{aligned} \left(1 + \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right) y_{t+1} - \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} y_t &= \left(1 + \frac{\sqrt{\kappa} + 1 - 2}{\sqrt{\kappa} + 1}\right) y_{t+1} - \frac{\sqrt{\kappa} + 1 - 2}{\sqrt{\kappa} + 1} y_t \\ &= \left(1 + 1 - 2 \cdot \frac{1}{\sqrt{\kappa} + 1}\right) y_{t+1} - \left(1 - 2 \cdot \frac{1}{\sqrt{\kappa} + 1}\right) y_t = (2 - 2\tau) y_{t+1} - (1 - 2\tau) y_t. \end{aligned}$$

Plugging into the expression for z_{t+1} from (5.28) gives

$$\begin{aligned} z_{t+1} &= \frac{1}{\tau} ((2 - 2\tau) y_{t+1} - (1 - 2\tau) y_t) - \frac{1 - \tau}{\tau} y_{t+1} \\ &= \frac{1}{\tau} ((2 - 2\tau) y_{t+1} - (1 - 2\tau) y_t - (1 - \tau) y_{t+1}) \\ &= \frac{1}{\tau} ((1 - \tau) y_{t+1} - (1 - 2\tau) y_t). \end{aligned}$$

Using the update rule (5.26) for y_{t+1} and the relation $x_t = (1 - \tau) y_t + \tau z_t$, we get

$$\begin{aligned} z_{t+1} &= \frac{1}{\tau} \left((1 - \tau) \left(x_t - \frac{1}{\beta} \nabla f(x_t) \right) - \frac{1 - 2\tau}{1 - \tau} (x_t - \tau z_t) \right) \\ &= \frac{1 - 2\tau}{1 - \tau} z_t + \frac{\overbrace{1 - 2\tau + \tau^2}^{=(1-\tau)^2} - 1 + 2\tau}{\tau(1 - \tau)} x_t - \frac{1 - \tau}{\tau\beta} \nabla f(x_t) \\ &= \frac{1 - \tau - \tau}{1 - \tau} z_t + \frac{\tau}{1 - \tau} x_t - \frac{1 - \tau}{\tau\beta} \nabla f(x_t) \\ &= \left(1 - \frac{\tau}{1 - \tau} \right) z_t + \frac{\tau}{1 - \tau} x_t - \frac{1 - \tau}{\tau\beta} \nabla f(x_t). \end{aligned}$$

Using $\tau = \frac{1}{\sqrt{\kappa} + 1}$ and $\beta = \kappa\alpha$, we get

$$\begin{aligned} z_{t+1} &= \left(1 - \frac{1}{\sqrt{\kappa} + 1} \cdot \frac{1}{1 - \frac{1}{\sqrt{\kappa} + 1}} \right) z_t + \frac{1}{\sqrt{\kappa} + 1} \cdot \frac{1}{1 - \frac{1}{\sqrt{\kappa} + 1}} x_t \\ &\quad - \left(1 - \frac{1}{\sqrt{\kappa} + 1} \right) \cdot (\sqrt{\kappa} + 1) \cdot \frac{1}{\kappa\alpha} \nabla f(x_t) = \left(1 - \frac{1}{\sqrt{\kappa} + 1} \cdot \frac{1}{\frac{\sqrt{\kappa} + 1 - 1}{\sqrt{\kappa} + 1}} \right) z_t \\ &\quad + \frac{1}{\sqrt{\kappa} + 1} \cdot \frac{1}{\frac{\sqrt{\kappa} + 1 - 1}{\sqrt{\kappa} + 1}} x_t - \left(\frac{\sqrt{\kappa} + 1 - 1}{\sqrt{\kappa} + 1} \right) \cdot (\sqrt{\kappa} + 1) \cdot \frac{1}{\kappa\alpha} \nabla f(x_t) \\ &= \left(1 - \frac{1}{\sqrt{\kappa} + 1} \cdot \frac{\sqrt{\kappa} + 1}{\sqrt{\kappa}} \right) z_t + \frac{1}{\sqrt{\kappa} + 1} \cdot \frac{\sqrt{\kappa} + 1}{\sqrt{\kappa}} x_t - \frac{\sqrt{\kappa}}{\alpha\kappa} \nabla f(x_t) \\ &= \left(1 - \frac{1}{\sqrt{\kappa}} \right) z_t + \frac{1}{\sqrt{\kappa}} x_t - \frac{1}{\alpha\sqrt{\kappa}} \nabla f(x_t). \end{aligned}$$

Hence,

$$\begin{aligned}
z_{t+1} - x^* &= \left(1 - \frac{1}{\sqrt{\kappa}}\right) z_t + \frac{1}{\sqrt{\kappa}} x_t - \frac{1}{\alpha\sqrt{\kappa}} \nabla f(x_t) - x^* \\
&= \left(1 - \frac{1}{\sqrt{\kappa}}\right) z_t + \frac{1}{\sqrt{\kappa}} x_t - \frac{1}{\alpha\sqrt{\kappa}} \nabla f(x_t) - \left(1 - \frac{1}{\sqrt{\kappa}}\right) x^* - \frac{1}{\sqrt{\kappa}} x^* - \frac{1}{\alpha\sqrt{\kappa}} \nabla f(x_t) \\
&= \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}}\right) (z_t - x^*) + \frac{1}{\sqrt{\kappa}} (x_t - x^*) - \frac{1}{\alpha\sqrt{\kappa}} \nabla f(x_t).
\end{aligned}$$

Using $\sqrt{\kappa} = \frac{1+\gamma}{\gamma}$ and $\sqrt{\kappa} - 1 = \frac{\gamma+1-\gamma}{\gamma} = \frac{1}{\gamma}$ (since $\sqrt{\kappa}\gamma = \frac{\sqrt{\kappa}}{\sqrt{\kappa}-1} = \frac{\sqrt{\kappa}-1+1}{\sqrt{\kappa}-1} = 1 + \frac{1}{\sqrt{\kappa}-1} = 1 + \gamma$), we get

$$\begin{aligned}
z_{t+1} - x^* &= \frac{1}{\gamma} \cdot \frac{\gamma}{1+\gamma} (z_t - x^*) + \frac{\gamma}{1+\gamma} (x_t - x^*) - \frac{\gamma}{\alpha(1+\gamma)} \nabla f(x_t) \\
&= \frac{1}{1+\gamma} (z_t - x^*) + \frac{\gamma}{1+\gamma} (x_t - x^*) - \frac{\gamma}{\alpha(1+\gamma)} \nabla f(x_t).
\end{aligned}$$

■

Theorem 5.6 (Accelerated GD: Well-Conditioned Optimization). *Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that is both α -strongly convex and β -smooth, define $\kappa := \beta/\alpha$. The update rules (5.26)-(5.27) ensure*

$$f(y_T) - f(x^*) \leq \exp(-T/\sqrt{\kappa}) \cdot \left(\frac{\alpha + \beta}{2} \|x_0 - x^*\|^2\right). \quad (5.29)$$

Proof. We assume $\kappa > 1$ for simplicity, since $\kappa = 1 \Leftrightarrow f(x) = ax^T x + b^T x + c$ for suitable scalars a, c and $b \in \mathbb{R}^n$. In this case, it is easily checked that the optimum solution is achieved in a single step. We define $\gamma = \frac{1}{\sqrt{\kappa}-1}$ and use the potential

$$\boxed{\phi_t = (1 + \gamma)^t \left(f(y_t) - f(x^*) + \frac{\alpha}{2} \|z_t - x^*\|^2\right)}.$$

The potential change is

$$\begin{aligned}
\phi_{t+1} - \phi_t &= (1 + \gamma)^{t+1} \left(f(y_{t+1}) - f(x^*) + \frac{\alpha}{2} \|z_{t+1} - x^*\|^2\right) \\
&\quad - (1 + \gamma)^t \left(f(y_t) - f(x^*) + \frac{\alpha}{2} \|z_t - x^*\|^2\right) \\
&= (1 + \gamma)^t ((1 + \gamma)(f(y_{t+1}) - f(x^*)) - (f(y_t) - f(x^*))) \\
&\quad + \frac{\alpha}{2} ((1 + \gamma) \|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2).
\end{aligned}$$

To show the convergence guarantee (5.29), it suffices to show that $\phi_{t+1} - \phi_t \leq 0$ for all t , which is equivalent to showing

$$(1 + \gamma)(f(y_{t+1}) - f(x^*)) - (f(y_t) - f(x^*)) + \frac{\alpha}{2} ((1 + \gamma) \|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2) \leq 0.$$

To bound the terms involving f , we use the smoothness of f with $x = x_t$ and $y = y_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$:

$$\begin{aligned} f(y_{t+1}) &\leq f(x_t) + \left\langle \nabla f(x_t), \underbrace{y_{t+1} - x_t}_{= -\nabla f(x_t)/\beta} \right\rangle + \frac{\beta}{2} \left\| \underbrace{y_{t+1} - x_t}_{= -\nabla f(x_t)/\beta} \right\|^2 \\ &= f(x_t) - \frac{1}{\beta} \|\nabla f(x_t)\|^2 + \frac{1}{2\beta} \|\nabla f(x_t)\|^2 = f(x_t) - \frac{1}{2\beta} \|\nabla f(x_t)\|^2. \end{aligned}$$

So,

$$\begin{aligned} (1 + \gamma)(f(y_{t+1}) - f(x^*)) - (f(y_t) - f(x^*)) \\ \leq f(x_t) - f(y_t) + \gamma(f(x_t) - f(x^*)) - (1 + \gamma) \frac{1}{2\beta} \|\nabla f(x_t)\|^2 \\ \leq \langle \nabla f(x_t), x_t - y_t \rangle + \gamma \left(\langle \nabla f(x_t), x_t - x^* \rangle - \frac{\alpha}{2} \|x_t - x^*\|^2 \right) - \frac{1 + \gamma}{2\beta} \|\nabla f(x_t)\|^2 \\ = \langle \nabla f(x_t), x_t - y_t + \gamma(x_t - x^*) \rangle - \frac{\alpha\gamma}{2} \|x_t - x^*\|^2 - \frac{1 + \gamma}{2\beta} \|\nabla f(x_t)\|^2. \quad (5.30) \end{aligned}$$

The last inequality uses convexity and strong convexity respectively. By definition (5.28),

$$z_t = \left(\frac{1}{\tau} - 1 \right) (x_t - y_t) + x_t \stackrel{\text{by definition of } \tau}{=} \sqrt{\kappa}(x_t - y_t) + x_t,$$

so we have $\gamma(z_t - x^*) = \sqrt{\kappa}\gamma(x_t - y_t) + \gamma(x_t - x^*)$. Using $\sqrt{\kappa}\gamma = 1 + \gamma$ (this equality holds since $\sqrt{\kappa}\gamma = \frac{\sqrt{\kappa}}{\sqrt{\kappa}-1} = \frac{\sqrt{\kappa}-1+1}{\sqrt{\kappa}-1} = 1 + \frac{1}{\sqrt{\kappa}-1} = 1 + \gamma$), we have $\gamma(z_t - x^*) = (1 + \gamma)(x_t - y_t) + \gamma(x_t - x^*)$. Dividing by $(1 + \gamma)$ gives

$$\frac{\gamma}{1 + \gamma} (z_t - x^*) = (x_t - y_t) + \frac{\gamma}{1 + \gamma} (x_t - x^*).$$

Using $\frac{\gamma}{1 + \gamma} = \gamma - \frac{\gamma^2}{1 + \gamma}$ for the term in RHS (this equality holds since $\frac{\gamma}{1 + \gamma} = \frac{\gamma + \gamma(1 + \gamma) - \gamma(1 + \gamma)}{1 + \gamma} = \gamma + \frac{\gamma(1 - 1 - \gamma)}{1 + \gamma} = \gamma - \frac{\gamma^2}{1 + \gamma}$), we get

$$(x_t - y_t) + \gamma(x_t - x^*) = \frac{1}{1 + \gamma} (\gamma(z_t - x^*) + \gamma^2(x_t - x^*)).$$

Substituting the above expression into (5.30) gives

$$\begin{aligned} (1 + \gamma)(f(y_{t+1}) - f(x^*)) - (f(y_t) - f(x^*)) \\ \leq \frac{1}{1 + \gamma} \langle \nabla f(x_t), \gamma(z_t - x^*) + \gamma^2(x_t - x^*) \rangle - \frac{\alpha\gamma}{2} \|x_t - x^*\|^2 - \frac{1 + \gamma}{2\beta} \|\nabla f(x_t)\|^2. \end{aligned} \quad (5.31)$$

To bound the terms in the potential change involving z , we use Claim 5.5 and Fact 1.2 with $a = z_t - x^*$, $b = \gamma(x_t - x^*)$ and $c = -\frac{\gamma}{\alpha}\nabla f(x_t)$

$$\begin{aligned}
& (1+\gamma) \|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2 \\
&= \frac{1}{1+\gamma} \left\| z_t - x^* + \gamma(x_t - x^*) - \frac{\gamma}{\alpha} \nabla f(x_t) \right\|^2 - \|z_t - x^*\|^2 \\
&= \frac{1}{1+\gamma} (\|z_t - x^*\|^2 + \gamma^2 \|x_t - x^*\|^2 + \frac{\gamma^2}{\alpha^2} \|\nabla f(x_t)\|^2 + 2\gamma \langle z_t - x^*, x_t - x^* \rangle \\
&\quad - \frac{2\gamma^2}{\alpha} \langle x_t - x^*, \nabla f(x_t) \rangle - \frac{2\gamma}{\alpha} \langle z_t - x^*, \nabla f(x_t) \rangle) - \|z_t - x^*\|^2 \\
&= \frac{1}{1+\gamma} (\|z_t - x^*\|^2 + \gamma^2 \|x_t - x^*\|^2 + \frac{\gamma^2}{\alpha^2} \|\nabla f(x_t)\|^2 + 2\gamma \langle z_t - x^*, x_t - x^* \rangle \\
&\quad - \frac{2\gamma^2}{\alpha} \langle \nabla f(x_t), x_t - x^* \rangle - \frac{2\gamma}{\alpha} \langle \nabla f(x_t), z_t - x^* \rangle) - \|z_t - x^*\|^2.
\end{aligned}$$

Now multiply this by $\frac{\alpha}{2}$ and add to 5.31 to get

$$\begin{aligned}
& (1+\gamma)(f(y_{t+1}) - f(x^*)) - (f(y_t) - f(x^*)) + \frac{\alpha}{2} ((1+\gamma) \|z_{t+1} - x^*\|^2 - \|z_t - x^*\|^2) \\
&\leq \frac{1}{1+\gamma} \langle \nabla f(x_t), \gamma(z_t - x^*) + \gamma^2(x_t - x^*) \rangle - \frac{\alpha\gamma}{2} \|x_t - x^*\|^2 - \frac{1+\gamma}{2\beta} \|\nabla f(x_t)\|^2 \\
&\quad - \frac{\alpha}{2(1+\gamma)} (\|z_t - x^*\|^2 + \gamma^2 \|x_t - x^*\|^2 + \frac{\gamma^2}{\alpha^2} \|\nabla f(x_t)\|^2 + 2\gamma \langle z_t - x^*, x_t - x^* \rangle \\
&\quad - \frac{2\gamma^2}{\alpha} \langle \nabla f(x_t), x_t - x^* \rangle - \frac{2\gamma}{\alpha} \langle \nabla f(x_t), z_t - x^* \rangle) - \frac{\alpha}{2} \|z_t - x^*\|^2 \\
&= \frac{\alpha}{2} \left(\frac{1}{1+\gamma} - 1 \right) \|z_t - x^*\|^2 + \frac{\alpha\gamma}{2} \left(\frac{\gamma}{1+\gamma} - 1 \right) \|x_t - x^*\|^2 + \frac{\alpha\gamma}{1+\gamma} \langle z_t - x^*, x_t - x^* \rangle \\
&\quad - \frac{1+\gamma}{2\beta} \|\nabla f(x_t)\|^2 + \frac{\gamma^2}{2\alpha(1+\gamma)} \|\nabla f(x_t)\|^2 + \frac{1}{1+\gamma} \langle \nabla f(x_t), \gamma(z_t - x^*) + \gamma^2(x_t - x^*) \rangle \\
&\quad - \frac{1}{1+\gamma} \langle \nabla f(x_t), \gamma(z_t - x^*) + \gamma^2(x_t - x^*) \rangle = \frac{\alpha}{2} \left(\frac{1}{1+\gamma} - 1 \right) \|z_t - x^*\|^2 \\
&\quad + \frac{\alpha\gamma}{2} \left(\frac{\gamma}{1+\gamma} - 1 \right) \|x_t - x^*\|^2 + \frac{\alpha\gamma}{1+\gamma} \langle z_t - x^*, x_t - x^* \rangle \\
&\quad + \frac{\alpha\gamma}{1+\gamma} \langle z_t - x^*, x_t - x^* \rangle - \frac{1+\gamma}{2\beta} \|\nabla f(x_t)\|^2 + \frac{\gamma^2}{2\alpha(1+\gamma)} \|\nabla f(x_t)\|^2.
\end{aligned}$$

The terms involving $\|\nabla f(x_t)\|^2$ cancel due to the equality $\frac{1+\gamma}{2\beta} = \frac{\gamma^2}{2\alpha(1+\gamma)}$, and this equality holds since

$$\frac{1+\gamma}{2\beta} = \frac{1+\gamma}{2\alpha \cdot \frac{\beta}{\alpha}} = \frac{1+\gamma}{2\alpha\kappa} = \frac{\frac{(1+\gamma)^2}{\kappa}}{2\alpha(1+\gamma)} \underbrace{\gamma = \frac{1+\gamma}{\sqrt{\kappa}}}_{=} \frac{\gamma^2}{2\alpha(1+\gamma)}.$$

Hence,

$$\begin{aligned}
& (1+\gamma)(f(y_{t+1})-f(x^*))-(f(y_t)-f(x^*))+\frac{\alpha}{2}((1+\gamma)\|z_{t+1}-x^*\|^2-\|z_t-x^*\|^2) \\
& \leq \frac{\alpha}{2}\left(\frac{1}{1+\gamma}-1\right)\|z_t-x^*\|^2+\frac{\alpha\gamma}{2}\left(\frac{\gamma}{1+\gamma}-1\right)\|x_t-x^*\|^2+\frac{\alpha\gamma}{1+\gamma}\langle z_t-x^*,x_t-x^*\rangle \\
& = \frac{\alpha}{2}\underbrace{\left(\frac{1-1-\gamma}{1+\gamma}\right)}_{=-\frac{\gamma}{1+\gamma}}\|z_t-x^*\|^2+\frac{\alpha\gamma}{2}\underbrace{\left(\frac{\gamma-1-\gamma}{1+\gamma}\right)}_{=-\frac{1}{1+\gamma}}\|x_t-x^*\|^2+\frac{\alpha\gamma}{1+\gamma}\langle z_t-x^*,x_t-x^*\rangle \\
& = -\frac{\alpha\gamma}{2(1+\gamma)}\left(\|z_t-x^*\|^2+\|x_t-x^*\|^2-2\langle z_t-x^*,x_t-x^*\rangle\right) \\
& \quad \underbrace{\text{using Fact 1.1 with } a=z_t-x^* \text{ and } b=-(x_t-x^*)}_{=}\quad -\frac{\alpha\gamma}{2(1+\gamma)}\|z_t-x^*-(x_t-x^*)\|^2 \\
& \quad = -\frac{\alpha\gamma}{2(1+\gamma)}\|z_t-x_t\|^2 \leq 0.
\end{aligned}$$

Hence, $\phi_{t+1}-\phi_t \leq 0$ for all t . We have

$$\phi_0 = (1+\gamma)^0 \left(f(y_0) - f(x^*) + \frac{\alpha}{2} \|z_0 - x^*\|^2 \right) = f(y_0) - f(x^*) + \frac{\alpha}{2} \|z_0 - x^*\|^2.$$

By β -smoothness of f with $y = y_0$ and $x = x^*$ and as $x_0 = y_0 = z_0$, we have

$$\begin{aligned}
f(y_0) - f(x^*) & \leq \underbrace{\langle \nabla f(x^*), y_0 - x^* \rangle}_{=0 \text{ (since } x^* \text{ is the minimizer and } \nabla f(x^*) = \vec{0})} + \frac{\beta}{2} \|y_0 - x^*\|^2 \\
& = \frac{\beta}{2} \|y_0 - x^*\|^2
\end{aligned}$$

and thus

$$\phi_0 \leq \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2. \quad (5.32)$$

By summing over t and using (5.32) we have

$$\begin{aligned}
\phi_T - \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 & \leq \phi_T - \phi_0 = \sum_{t=0}^{T-1} \phi_{t+1} - \phi_t \leq \sum_{t=0}^{T-1} 0 = 0 \\
& \Rightarrow \phi_T \leq \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2.
\end{aligned}$$

Hence, by definition of the potential:

$$\begin{aligned}
f(y_T) - f(x^*) &= (1 + \gamma)^{-T} \phi_T - \frac{\alpha}{2} \|z_T - x^*\|^2 \\
&\stackrel{\text{by removing the negative}}{\leq} \quad \text{squad-norm term} \quad (1 + \gamma)^{-T} \phi_T \leq (1 + \gamma)^{-T} \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 \\
&\Rightarrow f(y_T) - f(x^*) \leq (1 + \gamma)^{-T} \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 \\
&= \left(1 + \frac{1}{\sqrt{\kappa} - 1}\right)^{-T} \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 = \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} - 1} + \frac{1}{\sqrt{\kappa} - 1}\right)^{-T} \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 \\
&= \left(\frac{\sqrt{\kappa}}{\sqrt{\kappa} - 1}\right)^{-T} \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 = \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa}}\right)^T \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 \\
&= \left(1 - \frac{1}{\sqrt{\kappa}}\right)^T \frac{\alpha + \beta}{2} \|x_0 - x^*\|^2 \leq \exp(-T/\sqrt{\kappa}) \cdot \left(\frac{\alpha + \beta}{2} \|x_0 - x^*\|^2\right).
\end{aligned}$$

Hence the proof. ■

References

- [1] Bansal N. and Gupta A.: Potential-Function Proofs for Gradient Methods. In: Theory of Computing, 15(4), pp. 1-32 (2019).
- [2] Gupta A.: lecture notes of Advanced Algorithms course (Fall 2020) at CMU.
- [3] Takimoto E. and Warmuth M. K.: The Minimax Strategy for Gaussian Density Estimation. In: Proc. 13th Ann. Conf. on Computational Learning Theory (COLT'00), pp. 100-106. Morgan Kaufmann Publ., 2000.
- [4] Bubeck S.: Convex Optimization: Algorithms and Complexity. In: Foundations and Trends in Machine Learning, 8(3-4):231-357, 2015.
- [5] Nesterov Y.: A Method for Solving the Convex Programming Problem With Convergence Rate $O(1/k^2)$. In: Soviet Mathematics Doklady, 27:372-376, 1983.
- [6] Nesterov Y.: Introductory Lectures on Convex Optimization- A Basic Course. Kluwer Academic Publishers, 2004.
- [7] Hazan E.: Introduction to Online Convex Optimization, Second Edition. The MIT Press, 2019.

A Appendix

A.1 A Failed Attempt to Prove Theorem 3.12

As mentioned above, according to [1], Theorem 3.12 can be proved using Claim 3.11 and the same potential function used in the proof of Theorem 3.9. We assume $\kappa > 1$ for simplicity, since $\kappa = 1 \Leftrightarrow f(x) = ax^T x + b^T x + c$ for suitable scalars a, c and $b \in \mathbb{R}^n$. In this case, it is easily checked that the optimum solution x^* is achieved in a single step. Since no further details were given regarding the proof, it is implied that we have to follow the analysis from the proof of Theorem 3.9. I believe that otherwise more details would have been provided to prove Theorem 3.12. So we define $\gamma = \frac{1}{\kappa-1}$ and use the potential

$$\boxed{\phi_t = (1 + \gamma)^t \cdot (f(x_t) - f(x^*))}. \quad (\text{A.1})$$

The potential change is

$$\begin{aligned} \phi_{t+1} - \phi_t &= (1 + \gamma)^{t+1} \cdot (f(x_{t+1}) - f(x^*)) - (1 + \gamma)^t \cdot (f(x_t) - f(x^*)) \\ &= (1 + \gamma)^t \cdot ((1 + \gamma)(f(x_{t+1}) - f(x_t)) + \gamma(f(x_t) - f(x^*))). \end{aligned} \quad (\text{A.2})$$

As before, we need to show that $\phi_{t+1} - \phi_t \leq 0$ for all t . As we showed in the proof of Theorem 3.9, we have $-\frac{1+\gamma}{2\beta} + \frac{\gamma}{2\alpha} = 0$, so it is sufficient to showing

$$\phi_{t+1} - \phi_t \leq (1 + \gamma)^t \cdot \left(-\frac{1+\gamma}{2\beta} + \frac{\gamma}{2\alpha} \right) \|\nabla f(x_t)\|^2.$$

To do so, we need to show the following bounds for the two terms:

$$f(x_t) - f(x^*) \leq \frac{1}{2\alpha} \|\nabla f(x_t)\|^2$$

and

$$f(x_{t+1}) - f(x_t) \leq -\frac{1}{2\beta} \|\nabla f(x_t)\|^2$$

To bound the second term in (A.2), we use the strong convexity of f with $x = x_t$ and $y = x^*$:

$$f(x_t) - f(x^*) \leq \langle \nabla f(x_t), x_t - x^* \rangle - \frac{\alpha}{2} \|x_t - x^*\|^2 \leq \frac{1}{2\alpha} \|\nabla f(x_t)\|^2,$$

where the second inequality uses $\langle a, b \rangle - \frac{\|b\|^2}{2} \leq \frac{\|a\|^2}{2}$ with $a = \frac{1}{\sqrt{\alpha}} \nabla f(x_t)$ and $b = \sqrt{\alpha}(x_t - x^*)$. To bound the first term in (A.2), we use Claim 3.11 (as suggested) with $y = x_t$:

$$\begin{aligned} f(x_{t+1}) - f(x_t) &\leq \beta \langle x_t - x_{t+1}, x_t - x_t \rangle - \frac{\beta}{2} \|x_t - x_{t+1}\|^2 - \frac{\alpha}{2} \|x_t - x_t\|^2 \\ &= \underbrace{\beta \langle x_t - x_{t+1}, \vec{0} \rangle}_{=0} - \frac{\beta}{2} \|x_t - x_{t+1}\|^2 - \underbrace{\frac{\alpha}{2} \|\vec{0}\|^2}_{=0} \\ &= -\frac{\beta}{2} \|x_t - x_{t+1}\|^2. \end{aligned}$$

Using (3.13), we have

$$\begin{aligned} -\frac{\beta}{2} \|x_t - \tilde{x}_{t+1}\|^2 &\stackrel{(3.13)}{=} -\frac{\beta}{2} \left\| \frac{1}{\beta} \nabla f(x_t) \right\|^2 \\ &= -\frac{\beta}{2} \cdot \frac{1}{\beta^2} \|\nabla f(x_t)\|^2 = -\frac{1}{2\beta} \|\nabla f(x_t)\|^2. \end{aligned}$$

However, using the second part of Proposition 1.6 with $a = x_t$, $b = x_{t+1}$ and $\tilde{b} = \tilde{x}_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$, we get $\|x_t - x_{t+1}\|^2 \leq \|x_t - \tilde{x}_{t+1}\|^2$, so

$$-\frac{\beta}{2} \|x_t - x_{t+1}\|^2 \geq -\frac{\beta}{2} \|x_t - \tilde{x}_{t+1}\|^2 = -\frac{1}{2\beta} \|\nabla f(x_t)\|^2$$

and we cannot conclude that

$$f(x_{t+1}) - f(x_t) \leq -\frac{1}{2\beta} \|\nabla f(x_t)\|^2.$$

Therefore, in this way it is not possible to prove Theorem 3.12.